COMPARATIVE ANALYSIS OF OBJECT DETECTION ALGORITHMS FOR IMAGE-BASED DETECTION OF CHICKEN FOWLPOX DISEASE

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Bachelor of Science in Computer Science
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APPROVAL SHEET

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The Researchers

DEDICATION

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ABSTRACT

This study provides a Comparative Analysis of Object Detection Algorithms for Image-Based Detection of Chicken Fowlpox Disease namely: YOLOv8, Faster R-CNN, and RetinaNet for detecting Fowlpox lesions in chickens using annotated image datasets. A total of 1,275 labeled images of healthy and infected chickens were collected from open-source platforms and preprocessed using resizing, grayscale conversion, normalization, and data augmentation. The models were trained under consistent conditions and evaluated using Precision, Recall, F1-Score, and mean Average Precision (mAP@0.5 and mAP@0.5–0.95). YOLOv8 demonstrated superior performance with a Precision of 93.78%, Recall of 93.30%, F1-Score of 90.47%, mAP@0.5 of 95.92%, and mAP@0.5–0.95 of 73.29%. Faster R-CNN recorded a Precision of 89.75%, Recall of 79.11%, F1-Score of 84.11%, mAP@0.5 of 89.79%, and mAP@0.5–0.95 of 71.78%, while RetinaNet obtained a Precision of 92.82%, Recall of 77., 66%F1-Score of 84.57%, mAP@0.5 of 92.82%, and mAP@0.5–0.95 of 72.20%. Model validation against poultry expert evaluations was performed using a paired sample t-test. The results showed no statistically significant difference between the expert and the YOLOv8 model ($t(99) = -0.300$, $p = 0.765$), indicating strong agreement. However, RetinaNet displayed a significant difference ($t(99) = -5.366$, $p < 0.001$), while Faster R-CNN showed no significant deviation ($t(99) = 0.773$, $p = 0.441$). These outcomes validate YOLOv8's reliability for real-time automated poultry health monitoring, affirming its practical application for precision livestock farming.

Keywords*: Deep Learning, You Only Look Once (YOLOv8), Faster R-CNN, RetinaNet, Fowlpox Detection, Object Detection,*

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DEFINITION OF TERMS

This section provides definitions for the key technical terms used throughout this study. These definitions are essential for understanding the methodologies, concepts, and components of the developed models utilized in the comparative analysis of object detection algorithms for image-based detection of chicken Fowlpox disease.

Chicken Disease	Refers to any illness or infection that affects domestic chickens which may be caused by viral, bacterial, or parasitic agents. In the context of this study, chicken disease specifically pertains to conditions that can be visually identified in images, including symptoms such as fowlpox lesions, respiratory distress, or other observable indicators.
Deep Learning	A subset of machine learning that utilizes multilayered neural networks, known as deep neural networks, to simulate the complex decision-making power of the human brain.
Datasets	The dataset refers to the collection of information used for analysis. It includes both the measurements of swine and images captured from above the swine. This data will be used to develop and train models for estimating the swine' weight using computer vision and deep learning techniques.
Data Splitting	The division of a dataset into separate training and validation sets to ensure that model training and evaluation are performed on distinct data subsets.
Faster R-CNN	An enhanced version of the Faster RCNN object detection model, designed to improve detection accuracy and efficiency. It builds upon the original Faster R-CNN by incorporating advancements in backbone networks and detection modules, making it a powerful tool for object detection tasks.

Google Colab	Google Colab is an online platform for writing and running Python code. It provides access to GPUs and TPUs, making it useful for training and testing the computer vision models needed to estimate swine weights from images and measurements.
Livestock Farming	The practice of raising and maintaining animals such as cattle, poultry, and swine to produce meat, dairy, eggs, and other materials like wool and leather.
RetinaNet	A computer vision model used to detect objects in images. It is known for handling small and hard to detect objects using a special feature pyramid structure for better accuracy.
Roboflow	Roboflow is a tool used to manage, annotate, and preprocess images for machine-learning models. It helps annotate or bounding boxes of the symptoms of the skin lesions of chicken and the process of preparing the sick chicken image dataset, making it easier to train and test the datasets for comparative analysis
YOLOv8	Developed by Ultralytics, it is a cuttingedge computer vision model used in this study for detecting keypoints on swine's backs in top-down images. These points help measure back length and heart girth to estimate the swine's weight. YOLOv8 also supports object recognition, classification, and segmentation tasks.

CHAPTER I

INTRODUCTION

Background of the Study

The poultry industry plays a vital role in global food production, providing a major source of protein and livelihood for millions of people. However, poultry diseases such as Fowlpox pose a serious threat to production efficiency, animal welfare, and economic stability (Cartoni et al., 2022). Fowlpox is a contagious viral disease caused by the *Avipoxvirus*, which affects both the skin and mucous membranes of chickens. It is commonly characterized by wart-like lesions on the comb, wattles, eyelids, and other unfeathered areas. (Umar, B. N., Adamu, J., Ahmad, M. T., Ahmad, K. H., Sada, A., & Orakpoghenor, O., 2021). When left undetected, it can spread rapidly among flocks, leading to decreased egg production, slow growth, and even mortality. Early detection and prevention are therefore crucial to minimizing the disease's impact and ensuring the health and productivity of poultry farms (OIE, 2022).

Traditionally, Fowlpox detection relies on manual inspection by poultry farmers or veterinarians, which is often time-consuming, subjective, and prone to human error. With the advancement of computer vision and deep learning technologies, automated image-based disease detection has emerged as a promising alternative for efficient, reliable, and non-invasive monitoring of chicken health. Object detection algorithms allow computers to identify and locate disease symptoms directly from images, enabling faster and more accurate diagnosis compared to traditional methods. Among the most widely used object detection algorithms are YOLO (You Only Look Once), Faster R-

CNN (Region-Based Convolutional Neural Network), and RetinaNet. Each algorithm offers unique advantages and trade-offs in terms of accuracy, speed, and computational efficiency. YOLO is known for its real-time performance and single-stage detection process, making it suitable for practical applications where quick identification is essential. Faster R-CNN, on the other hand, uses a two-stage detection pipeline that provides higher accuracy in complex scenarios but with slower inference time. RetinaNet introduces a focal loss function to address class imbalance problems, achieving a balance between detection accuracy and speed.

This study focuses on the comparative analysis of these three object detection algorithms YOLOv8, Faster R-CNN, and RetinaNet for the image-based detection of Chicken Fowlpox disease. By evaluating their performance in terms of precision, recall, F1-score, and mean average precision (mAP), this research aims to identify which algorithm provides the most effective and reliable detection performance for diagnosing Fowlpox from chicken images. The findings of this study will contribute to the development of intelligent disease monitoring systems that can assist poultry farmers in making informed decisions, reducing disease spread, and improving overall poultry health management.

Ultimately, this research promotes the integration of artificial intelligence in agriculture, specifically in poultry disease monitoring. Through automated object detection, early identification of Fowlpox can be achieved, leading to better disease control strategies, reduced economic losses, and enhanced poultry welfare.

Objectives of the Study

The primary objective of this study is to conduct a comprehensive comparative analysis of various object detection algorithms used in the image-based detection of chicken disease.

Specifically, the study aims to:

1. Prepare and augment a labeled dataset of fowlpox-infected chickens for model training and testing.
2. Implement and train selected object detection algorithms, namely YOLO, RetinaNet, and Faster R-CNN, for detecting chicken disease.
3. Evaluate and compare the performance of the implemented algorithms using standard metrics such as mAP, precision, recall, and F1-score.
4. Identify the most effective object detection algorithm for detection of chicken disease based on the comparative analysis.

Significance of the Study

This study is important because it focuses on detecting chicken fowlpox disease using image-based deep learning algorithms. By comparing YOLOv8, Faster R-CNN, and RetinaNet, the research helps find which algorithm performs best in identifying infected and healthy chickens. This makes disease detection faster and more accurate compared to traditional manual observation, which often takes time and depends on human judgment.

For **poultry farmers**, this study can help them monitor the health of their chickens more easily. Detection of fowlpox through an automated system can prevent the spread of the disease, reduce production losses, and improve the

overall quality of poultry products. It also helps farmers make better decisions in managing their flocks and maintaining a healthy farm environment.

For **researchers and students**, this study serves as a reference for future works related to image-based disease detection and artificial intelligence applications in agriculture. It provides insights into how different object detection algorithms perform and how they can be used to solve real-world farming problems.

For **the agricultural and veterinary sectors**, the study promotes the use of modern technology like AI and computer vision to enhance animal health monitoring. It supports the goal of developing smart farming systems that are efficient, accurate, and sustainable.

This study is significant because it proves that using deep learning algorithms can help improve disease detection in chickens. It contributes to the modernization of poultry farming and encourages the use of technology to ensure animal health and food safety.

Scope And Limitations of the Study

This study focuses on the comparative analysis of object detection algorithms specifically YOLOv8, Faster R-CNN, and RetinaNet for the image-based detection of Chicken Fowlpox disease. The main objective is to evaluate and compare the performance of these algorithms in accurately detecting and classifying chickens as either Healthy (Normal) or Infected (Fowlpox) based on visual features present in the images. The scope of the study is limited to the use of image datasets collected from online sources such as Roboflow and Kaggle. These images consist of both healthy and infected chickens and were

augmented to increase dataset diversity and improve model performance. The dataset was divided into training, validation, and testing sets to ensure proper model evaluation. The comparison focuses on the algorithms' accuracy, precision, recall, F1-score, and mean Average Precision (mAP) as performance metrics. Model training and validation were conducted using a controlled environment with uniform dataset sizes, parameters, and hardware resources to maintain fairness in comparison.

However, the study is limited in several aspects. It only considers visible external symptoms of Fowlpox and does not include other poultry diseases or non-visible infections. The study also depends on the quality and variety of the dataset; environmental factors such as lighting, background, and image resolution may affect detection accuracy. Additionally, real-time deployment and hardware optimization (e.g., for mobile or farm use) are beyond the scope of this research.

Despite these limitations, the study provides valuable insights into the strengths and weaknesses of different object detection algorithms for poultry disease detection, serving as a foundation for future improvements and real-world applications.

CHAPTER II

THEORETICAL FRAMEWORK OF THE STUDY

Review of Related Literature and Studies

This chapter presents the relevant literature and studies acquired from both local and international sources. Literature is from books, and professional journal articles. This foundation will inform the methodology, analysis, and interpretation of findings in the following sections, providing a solid theoretical groundwork for addressing research problems.

All the research literature and studies are cited through a thorough review by the researchers in connection with various ideas, concepts, generalization, conclusion and developments related to the study. Furthermore, the information in this chapter serves as a guide and helps the researchers understand details that correspond to and are like current research.

Agriculture Livestock

Livestock agriculture involves the raising and management of animals mainly to produce meat, milk, eggs, wool, and leather. According to *Air Emissions from Animal Feeding Operations* (2003), this sector also includes animals raised for recreational purposes such as riding and racing, as well as those used for draft work. This broader definition highlights that livestock agriculture is not limited to food production alone but also supports transportation, labor, and leisure activities, particularly in rural communities. Livestock farming contributes significantly to global agricultural output, accounting for approximately 40 percent of total agricultural production. According to the Food and Agriculture Organization (FAO, 2009), this

contribution underscores the sector's critical role in ensuring food security and sustaining the livelihoods of nearly one billion people worldwide. The rapid expansion of livestock production reflects increasing global demand for animal-based products driven by population growth, urbanization, and rising income levels.

The Philippines is primarily an agricultural nation, where both crop and animal production play a vital role in the national economy. According to the Philippine Statistics Authority (PSA, 2015; 2016), animal production provides a livelihood for approximately 2.2 million households across the country. Galang (2017) explained that for many rural families, animal production is either the primary source of income or an essential component of diversified farming systems. This indicates that livestock farming contributes not only to food supply but also to poverty reduction and rural employment. Furthermore, animal production was identified as one of the fastest-growing agricultural subsectors in 2015 and ranked as the highest employment-generating sector in 2016, emphasizing its economic importance and its role in sustaining rural communities.

According to the FAO (2003), the main products of the Philippine livestock industry include goats, water buffalo, cattle, swine, and poultry. The industry is divided into small-scale semi-subsistence systems, commonly referred to as backyard or smallholder farms, and large-scale commercial production systems. This classification is based on animal population size, where backyard farms typically maintain fewer than 20 adult animals or fewer than 41 young animals. Farms exceeding these limits are categorized as

commercial operations. This distinction is important because backyard farms often rely on traditional practices with limited access to technology, while commercial farms tend to adopt more intensive production systems. Both farm types contribute to meeting the country's increasing demand for meat, particularly pork. According to Statista (2024), pork consumption reached 1.53 million tons in 2022 and is projected to increase to 2.2 million tons by 2032, reflecting changing dietary preferences. The swine population reached 9.77 million in 2023, with Central Luzon, CALABARZON, Western Visayas, and Northern Mindanao serving as major production hubs. From April to June 2023, swine production increased to 422.72 thousand metric tons liveweight, indicating continued growth despite disease and biosecurity challenges.

Chicken Production

Chicken is one of the most important agricultural commodities in the Philippines due to its affordability, cultural acceptance, and relatively fast production cycle. According to industry observations, global disease outbreaks such as bovine spongiform encephalopathy (BSE) in Europe and avian influenza in China have contributed to increased consumer preference for poultry over red meat. This shift has resulted in sustained growth in poultry demand. With continuous population growth and increasing per capita consumption, the poultry industry must expand its annual production capacity not only to meet domestic demand but also to remain competitive in potential export markets.

Globally, chicken farming is often conducted under intensive production systems, where large numbers of birds are raised within confined spaces.

According to Zhuang et al. (2018), such high-density environments significantly increase the likelihood of disease transmission once pathogens enter a flock. Although commercial farms typically implement strict biosecurity measures, breaches can still occur through contaminated feed, equipment, vehicles, or farm personnel. Avian Influenza and Newcastle Disease are among the most severe poultry diseases due to their high contagiousness and economic impact. Understanding disease transmission pathways is therefore essential in developing cause-and-effect models that link management practices to disease outbreaks. The spread and severity of poultry diseases depend on factors such as housing systems, vaccination programs, and environmental conditions, highlighting the need for comprehensive disease prevention strategies.

Chicken Disease

Fowl pox is a viral disease affecting birds of all ages and is caused by a virus belonging to the genus *Avipoxvirus*. According to Chut et al. (2023), this virus is closely related to the pathogens responsible for smallpox and cowpox in humans. The disease spreads through direct contact with infected birds, contaminated surfaces, droppings, or scabs, and can also be transmitted mechanically by mosquitoes. The multiple transmission pathways make fowl pox difficult to control, particularly in backyard and open housing systems where vector exposure is high.

According to Zhu and Sun et al. (2021), gut health remains one of the most persistent challenges in broiler chicken production. The widespread use of antibiotics as growth promoters has historically improved feed efficiency and production performance. However, this practice has also resulted in antibiotic

residues in chicken meat, raising serious food safety and public health concerns. Consequently, many countries have implemented regulations banning or restricting antibiotic use in livestock production. While antibiotic-free production systems are considered a safer alternative, their implementation poses significant challenges, including increased disease susceptibility and management complexity. This situation highlights the need for improved disease detection and prevention strategies that reduce reliance on antibiotics.

Chicken production is expected to continue increasing in the coming decade, emphasizing the growing importance of disease management in ensuring food security. According to Blake et al. (2020), avian coccidiosis ranks among the most economically significant poultry diseases in the United Kingdom, while South Asia remains one of the most affected regions by veterinary diseases impacting low-income populations. These findings illustrate that poultry diseases pose both economic and social challenges, particularly in developing regions where livelihoods depend heavily on poultry farming.

According to Xu et al. (2023), poultry diseases remain a major constraint to the sustainable development of the broiler industry. Milosevic et al. (2019) emphasized that early disease diagnosis is critical for improving productivity and animal welfare. However, traditional monitoring methods rely heavily on manual observation, which is time-consuming and prone to human error (Ojo et al., 2022). This limitation often results in delayed diagnosis, allowing diseases to spread before intervention occurs.

He et al. (2022) and Nakrosis et al. (2023) reported that poultry health status is reflected in physiological, physical, and behavioral symptoms,

including abnormal vocalizations, elevated body temperature, and altered movement patterns. These observable indicators provide valuable information for disease detection. Symptom detection technologies enable continuous, non-invasive, and automated monitoring, which can support early warning systems. Campler et al. (2024) demonstrated that machine learning models using feature selection techniques can successfully predict infectious bronchitis risk in broilers, highlighting the potential of data-driven approaches in poultry health management.

Importance of Early Disease Detection in Livestock Animals

Early disease detection is a fundamental component of effective animal health management. According to Setyati et al. (2024), early detection improves treatment effectiveness and reduces the overall impact of disease on individuals and society. Liu et al. (2020) reported that early diagnosis significantly reduces healthcare and treatment costs, while Ma' murutun et al. (2023) found that it lowers disease-related mortality rates. Advances in medical technology and improved understanding of disease mechanisms have further emphasized the importance of early detection (Arini & Syarli, 2020). Despite these benefits, Miller et al. (2019) noted that challenges such as limited access to technology and skilled personnel can hinder effective implementation. Nevertheless, Bronkhorst et al. (2019) emphasized that early diagnosis remains one of the most effective strategies for improving treatment outcomes.

According to Cowton et al. (2018), early disease detection in pigs is essential for minimizing economic losses and maintaining herd health, particularly for highly contagious diseases such as pleuropneumonia.

Opriessnig et al. (2011) explained that early identification enables timely treatment and reduces disease spread within herds. Kerr et al. (2003) reported that pleuropneumonia leads to high morbidity, mortality, reduced weight gain, and increased production costs. Although antibiotics are commonly used to manage outbreaks, Maes et al. (2018) warned that excessive use contributes to antimicrobial resistance. Therefore, early detection supports responsible antibiotic stewardship (Lekagul, 2019; Jorquera-Chavez et al., 2020).

Early and accurate diagnosis is considered the most effective approach to minimizing the economic burden of livestock diseases. According to Iotti et al. (2019) and Charfeddine and Pérez-Cabal (2023), subclinical infections result in hidden losses such as reduced feed efficiency, lower production yields, and premature culling. Hu et al. (2025) emphasized that delayed detection significantly increases financial losses, while Husnain and Usmani (2006) demonstrated that productivity losses due to morbidity often exceed losses from mortality.

According to the World Organisation for Animal Health (WOAH, 2024), early detection is the primary defense against epidemics and zoonotic diseases. Knight-Jones and Rushton (2013) emphasized that rapid detection is essential for controlling transboundary animal diseases that disrupt international trade. This aligns with the One Health approach, which recognizes that monitoring animal health is critical for preventing human health crises (FAO, 2023).

Technological Approaches to Disease Detection

According to Matthews et al. (2017), non-invasive disease detection methods such as monitoring behavior, vocalization, and skin temperature have

proven effective in pigs. These principles are transferable to poultry, where changes in movement, sound patterns, and thermal profiles can indicate disease. Sensor fusion techniques integrate multiple data sources, allowing for more comprehensive and reliable health assessments than single-sensor approaches. Continuous monitoring using sensors enables real-time disease detection and supports proactive intervention strategies.

Deep Learning Models for Disease Detection

According to Kamilaris and Prenafeta-Boldú (2018), artificial intelligence and computer vision technologies offer non-invasive solutions for precision livestock farming. CNNs are widely used due to their ability to extract hierarchical features from images (LeCun et al., 2015). Islam et al. (2021) demonstrated that CNNs combined with transfer learning can accurately detect poultry skin lesions even with limited datasets.

YOLO-based object detectors are widely adopted due to their real-time performance. Redmon et al. (2016) introduced YOLO, and subsequent versions have improved accuracy and speed. Elmessery et al. (2023) reported that YOLOv8 achieved exceptionally high accuracy in detecting poultry pathologies. Philippine-based studies by Evangelista et al. (2022) and Castañeda et al. (2022) further demonstrated YOLO's effectiveness under small dataset conditions.

RetinaNet, introduced by Lin et al. (2017), addressed class imbalance using Focal Loss and achieved accuracy comparable to two-stage detectors. Faster R-CNN remains a benchmark for accuracy despite its computational demands (Ren et al., 2015).

The literature shows that while many algorithms exist for image-based disease detection including CNN, ResNet, EfficientNet, and MobileNet the comparative evaluation of YOLOv8, Faster R-CNN, and RetinaNet is justified because these models represent diverse object detection strategies (onestage, two-stage, single stage detection). Each algorithm brings unique advantages that could enhance the detection of chicken diseases from small public datasets, thereby providing a comprehensive perspective on model suitability in real-world poultry health monitoring

Literature Matrix

Table 1. Matrix of Review of Related Literature

AUTHOR(S)	YEAR	TITLE	ALGORITHM / MODEL	FINDINGS
Campler et al.	2024	Predicting Infectious Bronchitis Broilers	Boruta Random Forest, ML	Applied machine learning to predict disease risks in broilers. Boruta Random Forest identified the most relevant features from farm data. Findings proved that ML models can complement deep learning by identifying risk factors before outbreaks occur.
FernandezCarol et al.	2024	Deep Learning for Poultry Disease Detection	CNN, ResNet	Focused on detecting poultry diseases through image datasets using CNN and ResNet. Results showed deep learning's high accuracy but highlighted the difficulty of acquiring large, labeled datasets. Recommended the use of online repositories and data augmentation.
Elmessery et al	2023	YOLO-based Model for Detecting Pathological Phenomena in Broiler Chickens	YOLOv8	Applied YOLOv8 to detect unhealthy broilers using augmented images. Achieved a high mAP of 0.988, proving the model's capability. The results showed YOLO's potential to support poultry farmers in early disease detection.

Li et al.	2023	Vision-Based Chicken Behavior Monitoring	CNN, RNN	Used CNN and RNN for behavior recognition in chickens. The model successfully identified stress-related behaviors that are linked to disease risks. Their study showed the importance of computer vision not only for direct disease detection but also for behavior monitoring.
Machuve et al.	2022	Poultry Disease Detection: A Review	DL review (YOLO, CNN, ResNet).	Reviewed deep learning models for poultry disease detection. Pointed out challenges such as limited annotated datasets, high costs of data collection, and lack of local datasets. The study suggested transfer learning and data augmentation as solutions.
Zhu & Sun et al.	2021	Gut Health Issues In Production Broiler	Analysis non (AI)	Analyzed how antibiotic use in poultry leads to residues in meat and resistance issues. Their findings emphasized the need for alternative monitoring strategies, such as AI-based detection of disease, to reduce reliance on antibiotics in chicken farms.
Singh & Kumar	2020	Comparative Study of YOLOv3, Faster R-CNN and SSD	YOLOv3, Faster R-CNN, SSD	Devices may be resource-limited. Compared three algorithms different on datasets. Found YOLOv3 was fastest, Faster R-CNN was most accurate, and SSD balanced both. The study is important for selecting algorithms depending on the tradeoff between speed and accuracy in poultry applications.
Zhou Homes et al.	2020	Abnormal Chicken Droppings Detection with Improved Faster R-CNN	Faster R-CNN	Developed an improved Faster R-CNN model for droppings detection in chickens. Reached 98.8% accuracy by using deformable convolution and feature enhancements. The study highlighted Faster R-CNN's strength in accuracy, though computationally slower than one-stage detectors.

Redmon et al.	2016	You Only Look Once: Unified Real-Time Object Detection	YOLO	Introduced YOLO as a one-stage detector that predicts bounding boxes and class probabilities directly. It achieved realtime performance without sacrificing much accuracy. The study influenced many later works on fast object detection in real-world applications.

Synthesis And Relevance Of The Study

The studies reviewed show that artificial intelligence and deep learning are very useful in detecting poultry diseases. Different algorithms like YOLO, Faster R-CNN, and CNN have been used by many researchers to identify if chickens are healthy or infected. For example, Elmessery et al. (2023) and Zhou Homes et al. (2020) showed that YOLO and Faster R-CNN can detect signs of sickness in chickens through images. Also, FernandezCarol et al. (2024) and Li et al. (2023) found that deep learning can also help observe chicken behaviors that may show stress or illness.

However, Machuve et al. (2022) said that one problem in using AI for poultry is the lack of enough image data and the high cost of collecting it. To solve this, they suggested using data augmentation and transfer learning to improve model results even with small datasets. Existing literature confirms the effectiveness of deep learning in automating poultry health monitoring, but notable gaps remain in its application. While YOLO, Faster R-CNN, and RetinaNet have each demonstrated strong results in object detection tasks, no comparative study has yet examined their performance on the same dataset specific to chicken Fowlpox detection.

This study addresses this gap by comparing the detection accuracy, precision, recall, and mAP@50 of three models: YOLOv8, Faster R-CNN, and RetinaNet, using a dataset of Fowlpox-infected and healthy chickens. The findings will help identify the most effective deep learning approach for real-time poultry disease detection beneficial to both small-scale and commercial Filipino poultry farmers.

CONCEPTUAL FRAMEWORK

Figure 1. Shows the conceptual framework of this study. The diagram illustrates the complete workflow for developing an image-based system that detects whether a chicken is healthy or infected with fowlpox. The process begins with data acquisition, where images of healthy and fowlpox-infected chickens are collected. These images undergo data preprocessing, which includes cleaning and labeling. After preprocessing, data augmentation is applied to increase the number of training samples by generating modified versions of the original images, such as rotated or flipped copies. Once the dataset is prepared, it is divided into training, validation, and testing sets to ensure proper learning and unbiased evaluation.

In the next stage, three object detection models Faster R-CNN, RetinaNet, and YOLOv8 are trained using the same grayscale dataset. Each model is designed to perform binary prediction, meaning it identifies whether the chicken in the image is “Healthy” or “Fowlpox,” along with a confidence score that indicates how certain the model is about its prediction. The use of grayscale images helps the model focus more on texture and pattern

differences rather than color, which is useful in detecting visible lesions on the skin of chickens.

To determine the best-performing model, several evaluation metrics are applied, including accuracy, precision, recall, F1 score, and mean average precision (mAP). These metrics provide a clear comparison of how well each model detects fowlpox cases and avoids incorrect classifications. After analyzing the results, a comparative assessment is conducted to identify the most effective model. The model with the highest performance is then selected as the final system for reliable and efficient fowlpox detection in chickens**.**

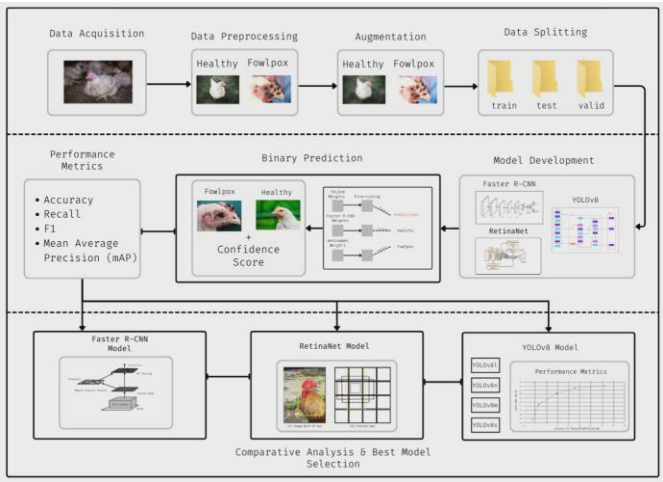


Figure 1. Conceptual Framework

CHAPTER III

METHODOLOGY

This chapter presents the methodological framework used to conduct a Comparative Analysis of Object Detection Algorithms: YOLOv8, Faster R-CNN and RetinaNet for Image-Based Detection of Fowlpox Chicken Disease. The primary objective of the study is to evaluate and compared the performance of each algorithm based on criteria such as accuracy, precision, recall, F1-score, and detection speed, to determine the most effective algorithm for detection of chicken diseases.

Study Area

This study was conducted using a combination of online data and local expert information to ensure both technical validity and real-world relevance. The primary dataset was sourced from Roboflow and Kaggle platform that hosts open-source image datasets used for computer vision tasks. It included images of both healthy chickens and those showing visible symptoms of Fowlpox. These images were used for the training, validation, and testing of the deep learning model developed in this study. The use of diverse and annotated online images provided a strong foundation for developing a reliable and generalizable model for disease detection.

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These images were used for the training, validation, and testing of the deep learning model developed in this study. The use of diverse and annotated online images provided a strong foundation for developing a reliable and generalizable model for disease detection.

In addition to digital data, the researchers also sought practical insights through an interview with a poultry expert from the City Veterinary and Slaughterhouse Management Department in Calamba City, Laguna. The interview helped the researchers understand the local practices, symptoms, and real-life conditions of Fowlpox infection among chickens in the area. This collaboration ensured that the study not only focused on model performance but also addressed how the model's predictions align with field observations and existing poultry management practices.

Thus, the study area represented using Roboflow datasets for model development, and a physical scope, signified by the inclusion of expert knowledge from Calamba City, Laguna. This combination allowed the researchers to integrate both artificial intelligence techniques and agricultural insights to produce a system relevant to modern poultry health monitoring.

Materials

The research was conducted using various hardware and software tools essential for the development and completion of the study. These materials enabled the implementation of comparative analysis

Software

This study used several software tools to prepare and analyze images for disease detection. These tools helped with labeling images, training the

model, testing its accuracy, and showing the results. Cloud platforms, coding environments, and machine learning libraries were used to support the process.

Google Colaboratory (Colab)

Colaboratory, or Colab, is a cloud-based platform provided by Google that allows users to run Python code in an interactive Jupyter Notebook environment with GPU support. In this study, Colab served as the main platform for training and evaluating object detection models. It was used to implement and train YOLOv8, Faster R-CNN, and RetinaNet using datasets imported from Roboflow. Colab's GPU resources made it convenient to handle model experimentation and performance comparison for the detection of visible symptoms of Fowlpox in chickens.

Roboflow

Roboflow is an online platform used for managing image datasets. In this study, it was used for annotating the images, applying data augmentation, and exporting the dataset into a format compatible with YOLOv8, Faster R-CNN, and RetinaNet. Roboflow's preprocessing tools, such as image resizing and brightness adjustment, helped prepare a clean and consistent dataset for training the models. While the annotations focused on detecting visible symptoms of Fowlpox in chickens, the platform supports future expansion for more detailed tagging.

Canva

Canva is an online graphic design platform used to create visual materials. In this study, Canva was used to design figures, charts, and presentation materials for visualizing model results and summarizing findings.

Its user-friendly interface allowed the researchers to create clear and professional graphics for the documentation and defense of the study.

Kaggle

Kaggle is an online platform that hosts a wide range of datasets for data science and machine learning projects. In this study, Kaggle was used to obtain additional images related to chicken diseases, which were included in the dataset used for model training and evaluation. The platform provided openly accessible and high-quality data that supported the expansion and diversity of the dataset.

Python

Served as the main programming language for the development of the object detection model. It was chosen for its simplicity, flexibility, and compatibility with various deep learning frameworks and libraries.

Ultralytics YOLOv8

A high-performance object detection framework based on the YOLO (You Only Look Once) architecture. It was used for training, validation, and inference of the YOLOv8 model. The framework includes built-in support for dataset handling, augmentation, and visualization.

Torchvision

Torchvision is a key library within the PyTorch ecosystem, widely used for computer vision tasks. In this study, Torchvision played a crucial role by providing pre-trained models such as Faster R-CNN and RetinaNet, which were essential for comparative analysis against YOLOv8. These models were fine-tuned to detect visible symptoms of Fowlpox in chickens. The library also offered a collection of utility functions for data augmentation and image

transformation, including resizing, normalization, and flipping of images, which helped enhance the robustness of the input data.

OpenCV

OpenCV (Open-Source Computer Vision Library) was utilized extensively for image preprocessing, augmentation, and visualization of results. It enabled various image operations such as resizing, grayscale conversion, brightness and contrast adjustment, and noise reduction. OpenCV was also used to draw bounding boxes on images, helping to visually assess the accuracy of model predictions during both training and testing phases. This made it a valuable tool for both data preparation and post-training evaluation.

NumPy

NumPy is a fundamental Python library for numerical computing, which plays an essential role in handling data arrays and numerical operations throughout the research. It was used for efficiently managing image tensors, manipulating label arrays, and performing mathematical operations needed during both preprocessing and evaluation. Its seamless integration with other libraries like PyTorch and OpenCV ensured smooth data flow during the model development pipeline.

Matplotlib

Matplotlib was employed for visualizing the performance of object detection models across the experiments. It was used to generate plots for training and validation of loss, accuracy trends, and confusion matrices. Comparative visualizations were also created to showcase the performance differences between YOLOv8, Faster R-CNN, and RetinaNet. These visual

insights were crucial for analyzing model behavior and making informed conclusions.

Pandas

Pandas were utilized for organizing and analyzing performance metrics obtained during model evaluation. It helped structure experimental data in tabular form, enabling straightforward manipulation and comparison of precision, recall, F1 scores, and mAP values across the three models. Its data-handling capabilities supported the efficient summarization of results, which contributed significantly to the study's comparative analysis.

Hardware

The model in this study was developed, tested, and implemented on a local machine with the following specifications.

Computer

The computer served as a personal workstation for building and testing the object detection algorithms used in comparative analysis. It provided the necessary processing power and resources required for training YOLOv8, Faster R-CNN, and RetinaNet, as well as handling the dataset. Table 2. Presents the machine specifications that were utilized during the development and evaluation of the comparative detection models.

Table 2. Desktop Specification

DESKTOP MINIMUM SPECIFICATION	
CPU	Intel® Core™ i7-4600M Processor
RAM	8.0 Gigabiytes
GPU	Integrated Intel® HD Graphics 4600
STORAGE	256 Gigabytes HDD/SSD

Research Design

This study utilized a developmental and experimental research design with a comparative approach to evaluate the performance of three object detection algorithms YOLOv8, Faster R-CNN, and RetinaNet in detecting Fowlpox disease in chickens. The developmental component focused on the design and construction of an AI-based object detection system for Fowlpox detection. The experimental component involved implementing, training, and testing the three algorithms under controlled conditions to compare their performance.

The study involved collecting images of chickens exhibiting visible Fowlpox symptoms as well as healthy chickens from online sources and selected poultry farms for healthy chickens. These images were annotated and preprocessed to create a labeled dataset suitable for model training. The three algorithms were then implemented and trained using the same dataset to ensure a fair and consistent comparison. A portion of the dataset was reserved for testing to validate each model's performance on unseen images.

The outputs of the algorithms were evaluated using standard object detection performance metrics, including precision, recall, F1-score, and mean average precision (mAP), allowing for a systematic comparison of their detection accuracy and efficiency. By employing this developmental and experimental research design, the study provided empirical evidence on the effectiveness of each algorithm, supporting the development of reliable AI-based systems for Fowlpox detection and poultry health monitoring.

Model Development

The development of the model for detecting skin lesions caused by Fowlpox disease in chickens involved a structured process consisting of data acquisition, data preprocessing, model training, and model evaluation. This process ensured that the image-based detection models were accurately trained and assessed using comparable parameters across the three selected algorithms: YOLOv8, Faster R-CNN, and RetinaNet.

The dataset used in this study consisted of annotated images of chickens showing visible skin lesions associated with Fowlpox infection and healthy chicken without any lesions. Each image was preprocessed through normalization, resizing, and augmentation techniques such as flipping, rotation, and brightness adjustment. These preprocessing steps helped increase the dataset's diversity and improved model generalization to varying image conditions. The dataset was divided into training, validation, and testing sets to ensure fair model assessment and prevent bias.

The YOLOv8 (You Only Look Once version 8) algorithm served as a one-stage object detector known for its balance between detection accuracy and real-time performance. It directly predicts bounding boxes and class probabilities in a single forward pass, making it efficient for detecting multiple lesions in a single image. The Faster R-CNN (Region-Based Convolutional Neural Network) model, on the other hand, operates as a two-stage detector. It first proposes regions of interest (ROIs) through a Region Proposal Network (RPN) and then performs classification and bounding box regression, providing high precision but requiring more computation time. Meanwhile, the RetinaNet

model employs a single-stage detection framework that uses Focal Loss to address class imbalance, making it effective in detecting small and less distinct lesion areas commonly found in infected chickens.

During the model training phase, each algorithm was trained using identical experimental settings to maintain consistency in comparison. Key hyperparameters, including learning rate, number of epochs, batch size, optimizer, etc. were adjusted to achieve optimal convergence and detection accuracy. The training process focused on enabling each model to learn the distinctive texture, color, and shape patterns of Fowlpox lesions on chicken skin. The model evaluation phase involved assessing the trained models based on standard object detection performance metrics: Precision, Recall, mean Average Precision (mAP), and F1 Score. These metrics were used to determine how effectively each model could identify and localize skin lesions. Additionally, confusion matrices and precision-recall (PR) curves were utilized to visualize the trade-offs between correctly detected and missed lesions. Through these evaluations, the comparative performance of YOLOv8, Faster R-CNN, and RetinaNet was determined, highlighting which algorithm demonstrated the most accurate and reliable detection of Fowlpox skin lesions in chickens.

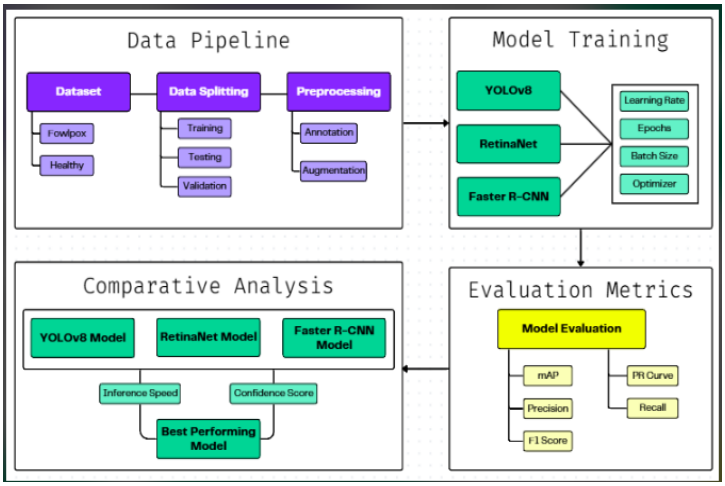


Figure 2. Model Development

Data Acquisition

The study will gather data through image collection of chickens showing visible symptoms of Fowlpox disease, as well as images of healthy chickens, to ensure that the model can accurately distinguish between infected and non-infected conditions. The primary dataset will be obtained from Roboflow, an open-source platform that provides annotated images for computer vision research. Additionally, the researchers will collect supplementary image data from local poultry farms using mobile phone cameras to capture real-world scenarios under natural lighting and environmental conditions. These images will focus on identifying external signs of Fowlpox, such as scabs, lesions, or nodular growths on the chicken comb, wattles, and skin.

The dataset will serve as the foundation for training and testing the object detection models used in this study. The collected images will undergo preprocessing steps such as resizing, enhancement, and augmentation techniques including rotation, flipping, and scaling to increase dataset variability

and model robustness. The data will then be divided into training, validation, and testing sets, which will be used to develop and evaluate the models. Through this process, the study aims to generate a well-structured and high-quality dataset that supports reliable model performance for image-based detection of Fowlpox disease in chickens.

Data Sources

Table 3. Shows the dataset used in this study consists of images of healthy chickens captured using a mobile phone at a local poultry farm. These images were carefully collected and annotated to train machine learning algorithms for detecting and identifying chickens in various environmental conditions. The use of healthy chicken images serves as a baseline for understanding normal physical appearance and behavior, which can later be compared to disease-affected samples in future research related to Fowlpox detection. Each image was manually annotated with bounding boxes using a two classes label: fowl-pox chicken and healthy chicken." The images were captured and collected under mixed lighting conditions, including both natural daylight and artificial lighting, to ensure data diversity and improve the model's adaptability. These annotated images form the foundation for training, validating, and testing the object detection models used in this study.

In addition to image collection, interviews were conducted with the owner of a small backyard farm and the City Veterinary Office of Calamba, Laguna. These interviews aimed to gather information regarding Fowlpox disease, its symptoms, transmission, and common prevention and control practices. The insights from these interviews helped the researchers better understand the

disease's impact on poultry production and provided essential context for the study.

Table 3. Data Description

Data Type	Images (JPEG)
Source	Online Sources (kaggle, Roboflow)
Chicken Status	chicken with fowlpox and healthy
Number of Images	1,275
Image Resolution	640x640
Annotation	Bounding box
Environmental Condition	Mixed natural and artificial lighting
Additional Data	Interviews with small poultry backyard owner and Calamba City Veterinary Office

Table 4. Shows that the dataset is divided into three subsets to ensure effective training, validation, and testing of the object detection models. A total of 1,275 images were used in the study. The training set, comprising 75% of the dataset or 954 images, is used to teach the model to recognize and detect visible symptoms of Fowlpox in chickens under various conditions. The validation set, consisting of 12% or 160 images, is utilized during training to fine-tune hyperparameters and monitor performance, helping to prevent overfitting. The remaining 13% or 161 images form the test set, which is held out during training and used exclusively to evaluate the final performance of the trained models. This structured division ensures that the models undergo rigorous training and reliable evaluation, enhancing their ability to generalize well to unseen data.

Table 4. Data Splitting

Datasets	Number of Samples	Percentage
Training	954	75%
Testing	161	13%
Validation	160	12%
Total	1,275	100%

Model Selection

The model selection process for this study was based on the goal of accurately detecting Fowlpox skin lesions in chickens using image-based object detection techniques. Given the nature of the dataset, which consists of annotated images showing visible lesions and normal skin areas, three state-of-the-art object detection algorithms were selected for comparative analysis: YOLOv8, Faster R-CNN, and RetinaNet. These models were chosen based on their proven performance in object localization, accuracy, and efficiency across diverse image detection tasks. The first selected algorithm, YOLOv8 (You Only Look Once version 8), is a one-stage real-time object detector known for its speed and high accuracy. It uses an advanced convolutional backbone and detection head that simultaneously predicts bounding boxes and class probabilities in a single network pass. This design makes YOLOv8 highly efficient for detecting multiple lesions within a single image, even under varied lighting and environmental conditions. Its real-time capability and balanced precision make it suitable for fast and accurate lesion detection tasks.

The second algorithm, Faster R-CNN (Region-Based Convolutional Neural Network), represents a two-stage detection framework. It first generates region proposals through a Region Proposal Network (RPN), which identifies potential areas of interest, and then performs classification and bounding box

regression on these regions. Faster R-CNN is known for its high detection of precision and robust feature extraction, making it particularly effective for identifying small or complex lesions that require detailed spatial analysis. Although slower than YOLOv8, its accuracy in handling fine-grained detection tasks makes it an ideal comparison model.

The third algorithm, RetinaNet, combines the speed of single-stage detectors with improved accuracy through the introduction of the Focal Loss function. This function addresses the issue of class imbalance by reducing the loss of contribution from easy-to-detect samples and focusing on hard, misclassified examples. RetinaNet is especially useful for detecting subtle or partially visible skin lesions that might be overlooked by traditional models. Its balance of efficiency and sensitivity makes it a strong candidate for lesion detection in poultry disease studies. By employing these three algorithms, the study aims to compare their performance in terms of detection of accuracy, precision, recall, and mean Average Precision (mAP). The comparative analysis identifies which algorithm demonstrates the best balance between accuracy and computational efficiency for detecting Fowlpox skin lesions in chickens. This selection process ensures that the study adopts a scientifically grounded approach in determining the most effective model for image-based poultry disease detection.

Model Evaluation

The evaluation of the models focused on measuring their ability to accurately detect Fowlpox lesions in chickens. Three object detection algorithms YOLOv8, Faster R-CNN, and RetinaNet. After training, the models

were evaluated using standard performance metrics, including Precision, Recall, F1-Score, and Mean Average Precision (mAP@0.5). These metrics measured how accurately and consistently each model detected Fowlpox lesions. Additional tools such as Precision-Recall (PR) curves and confusion matrices were used to analyze model performance in detail.

To ensure fairness and accuracy, the training and validation losses were monitored throughout the process to check for overfitting and verify stable learning. This evaluation helped identify which algorithm provided the best performance balance between detection accuracy and processing speed. Through this structured development process, the study successfully compared the performance of YOLOv8, Faster R-CNN, and RetinaNet, identifying their strengths and weaknesses in detecting chicken Fowlpox lesions. The results provided insights into which model is most suitable for reliable and practical use in poultry disease detection.

Precision

Measures the proportion of True positive instances among those classified as positive, is also called Positive Predictive Value. Higher precision indicates fewer false positive detections, implying that the algorithm is more accurate at identifying objects. Precision is calculated as the ratio of true positive predictions to the total positive predictions (true positives plus false positives). It measures the model's ability to correctly identify fowl pox disease.

Recall

Recall, alternatively termed sensitivity or True Positive Rate, is another important metric used in evaluating object detection models. The model's ability

is measured to identify positive instances in comparison to actual positives (Abad & Castro, 2016).

The formula calculates the ratio of correctly identified positive instances (TP) to all actual positive instances (TP + FN). Here, True Positives (TP) refer to detected chicken disease, while False Negatives (FN) are the chicken the model missed.

F1 Score

The F1 score measures a model's overall performance. It considers both precision (how many of its guesses are correct) and recall (how many correct things it finds). The F1 score balances these two factors to assess the model's performance fairly. The formula computes the harmonic mean of precision and recall, providing a balanced score even when precision and recall differ.

Mean Average Precision (mAP)

For object detection algorithms like YOLO, Faster R-CNN and RetinaNet mAP are used to evaluate the models (Shah D., 2022). The score is calculated by comparing the detected box with the ground-truth bounding box. A higher mAP score indicates more accurate detection and prediction by the model. The mean Average Precision (mAP) is calculated by finding each class's Average Precision (AP) and averaging it across multiple classes. It comprehensively measures the model's performance in detecting and retrieving image data in this study.

After all evaluation metrics were computed, the performances of YOLOv8, Faster R-CNN, and RetinaNet were compared to determine which

model performed best overall. The comparison focused on accuracy-based metrics such as mean Average Precision (mAP), Precision, Recall, and F1score. Through this analysis, the most effective model for Fowlpox detection was identified based on its high accuracy and strong overall performance.

Model Validation

To ensure the reliability and accuracy of the object detection models used in this study, a T-test validation was conducted. This statistical test was employed to determine whether the differences in the performance of the three models YOLOv8, Faster R-CNN, and RetinaNet were statistically significant when applied to the validation dataset.

A subset of the collected dataset, consisting of images that were not used during training, was set aside specifically for model validation. Each image in this subset was annotated to indicate whether the chicken was healthy or exhibited visible Fowlpox lesions. The models were then applied to these validation images, and their detection results including predicted labels, confidence scores, and identified lesions.

By applying T-test validation, the study ensured that the selection of the best-performing object detection model was supported not only by observed accuracy but also by rigorous statistical evidence. This approach strengthens the credibility of the findings and provides a reliable basis for recommending the most effective model for automated Fowlpox detection in poultry farms

Ethical Consideration

The researchers ensured that this study adhered to ethical standards in conducting research involving image data and artificial intelligence systems. The dataset utilized in this research was composed exclusively of publicly available images from trusted online platforms such as Roboflow and Kaggle. All images were used in accordance with their respective usage licenses, and no proprietary or unauthorized datasets were accessed or distributed. Additionally, the dataset did not contain any sensitive, personal, or identifiable information about individuals.

No live animals were directly involved in this study. The research solely relied on digital images and therefore did not involve any form of animal handling, experimentation, or intervention. This ensured full compliance with animal welfare guidelines and avoided any chance of causing harm or distress to chickens or other living creatures. All images showing disease symptoms were repurposed responsibly for academic and research purposes alone. The researchers recognize the responsibility associated with applying artificial intelligence in fields such as agriculture and animal health. While the study is geared toward advancing technological solutions for early disease detection, the researchers acknowledge that AI predictions are not a substitute for veterinary expertise. Misdiagnosis could lead to improper treatment actions, unnecessary culling, or economic losses for farmers. Therefore, the object detection models developed in this study are intended only as supportive tools that require professional verification before being applied in real-world scenarios.

Lastly, the researchers ensured transparency and integrity throughout the study. All findings, whether favorable or not, were reported honestly without manipulation. The use of comparative analysis also aimed to identify strengths and limitations of different models in a fair and unbiased manner, contributing to the shared knowledge in the field of computer vision and poultry disease monitoring.

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents the results and analysis of the three object detection algorithms YOLOv8, Faster R-CNN, and RetinaNet used for detecting Fowlpox disease in chickens. The results include model training outcomes, performance evaluations, visual detection outputs, and a comparative discussion based on accuracy, precision, recall, and mAP scores.

Data Gathered

The data collection for this study, titled “Comparative Analysis of Object Detection Algorithms for Image-Based Detection of Chicken Fowlpox Disease,” focused on gathering images of chickens that are either healthy or infected with Fowlpox. These images were used to train and test the object detection models: YOLOv8, Faster R-CNN, and RetinaNet.

The images were collected from online sources such as Roboflow and Kaggle. These websites provide free and open datasets that include different poultry images suitable for research. The images show visible symptoms of Chicken Fowlpox, such as lesions or scabs on the comb, wattles, and eyelids, while others show healthy chickens for comparison.

A total of 639 original images were collected. To increase the amount of data and make the model more reliable, data augmentation was applied. This includes rotating, flipping, adjusting brightness and contrast, and adding noise to the images. After augmentation, the dataset grew to 1,275 images in total. The dataset was divided into three parts: 75% for training (954 images), 12% for validation (160 images), and 13% for testing (161 images). This division

helped make sure that all three models were tested fairly using the same data. Each image was also annotated with bounding boxes to mark the infected parts of the chicken. The collected and processed images became the main foundation of this study. These images provided enough examples for the models to learn how to identify and detect signs of Chicken Fowlpox accurately.

Figure 3 shows two types of chicken conditions used in the study. Image (a) represents a healthy chicken, which shows no visible signs of infection or abnormal skin growth. Its comb and wattle appear smooth, clean, and evenly colored, indicating that the chicken is in normal condition. This type of image is used as a reference for detecting what a non-infected chicken looks like, helping the model learn the features of a healthy class. In contrast, image (b) shows a chicken infected with Fowlpox, a viral disease that causes wart-like lesions on the comb, wattles, eyelids, and other unfeathered parts of the body. The raised, rough, and swollen bumps seen on the chicken comb are classic symptoms of the disease. These visible lesions allow the detection model to distinguish between healthy and infected chickens based on texture and skin abnormalities. By comparing normal and infected samples, the model can learn to identify Fowlpox more accurately.



(a) (b)

Figure 3. Sample images used in the dataset: (a) Healthy chicken (b) Fowlpox-infected chicken.

Data Preparation and Augmentation

Upon the initial stage of model development, the researchers collected multiple images of chickens from various online sources, including Roboflow and Kaggle. These images included both healthy and Fowlpox-infected chickens. Each image was carefully examined to ensure that visible symptoms such as lesions and scabs on the comb, wattles, and eyelids. After collecting the raw images, the researchers uploaded the dataset to Roboflow for manual annotation. Each image was labeled using bounding boxes to mark the infected areas of the chickens' skin, ensuring that the models could learn to detect and localize Fowlpox symptoms accurately.

Figure 4 shows a sample portion of the raw dataset containing Fowlpox-infected chickens, which was used in the development of the detection model. Each image displays visible symptoms of Fowlpox, such as wart-like lesions, scabs, and swelling on the comb, wattles, and facial area of the chicken. The dataset includes a wide variety of image conditions, such as different angles,

lighting environments, distances, and backgrounds. This variation ensures that the model learns to recognize Fowlpox in both clear and imperfect real-world scenarios, rather than relying on uniform or ideal images.

This dataset was annotated and prepared for training, allowing the model to identify the disease by learning the visual patterns present in the infected samples. The diversity of images helps prevent overfitting and improves the model's ability to generalize when analyzing new, unseen data. By using real and unfiltered images gathered from various online sources, the dataset provides realistic examples of how Fowlpox appears in actual poultry conditions, making the model practical and applicable to real-world poultry disease monitoring.

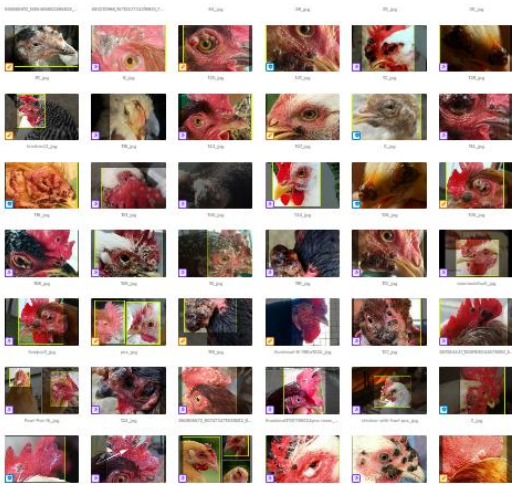


Figure 4. Annotation of Chickens with Fowlpox Lesions in Roboflow

Figure 5 shows the preprocessing settings indicate the steps applied to each image before it was used for training the model. First, the images were automatically oriented to ensure they were properly aligned. Then, all images were re-sized to 640×640 pixels to maintain a consistent input size for the model. Adaptive contrast adjustment was applied to enhance the visibility of key features, making the lesions and skin abnormalities clearer for the model to detect. The augmentation settings show how additional training samples were generated to improve model performance and reduce overfitting. For every original image, three augmented versions were produced by randomly adjusting visual properties such as hue, saturation, brightness, and exposure. Minor blur and noise were also added to simulate real-world image imperfections. These variations help the model learn to recognize Fowlpox even under different lighting, camera quality, and environmental conditions, making it more reliable when applied outside controlled settings

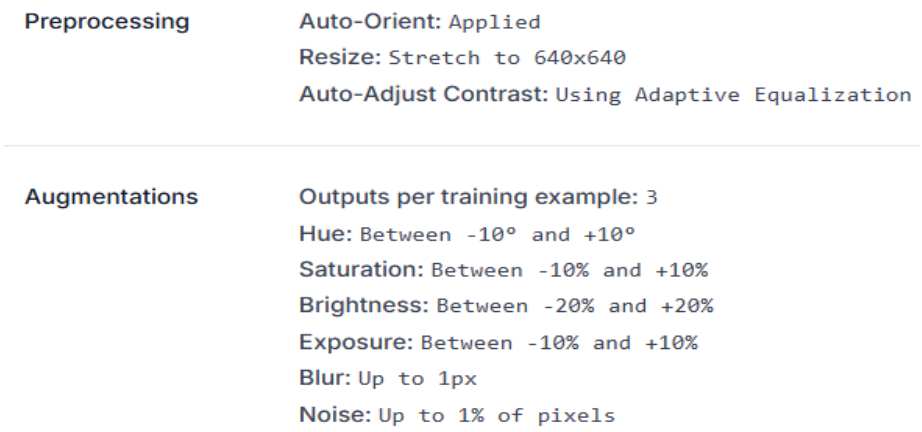


Figure 5. Image Preprocessing and Augmentations

Model Training Parameters

This section presents the hyperparameter configurations used in training the three object detection algorithms. The tuning process involved adjusting values such as batch size, learning rate, number of epochs, and optimizer type to achieve optimal model accuracy.

The table 5 shows the main hyperparameters used during the training of the YOLOv8 model. The model used is Yolov8s.pt, which is a small but efficient version of YOLOv8. It was trained for 40 epochs with a batch size of 16, meaning 16 images were processed at a time during training. The image size (imgsz) was set to 640, which defines the resolution of the input images. The learning rate of 0.01 controlled how fast or slow the model learned during training, while the optimizer was set to Auto to automatically select the best optimization method. The momentum value of 0.937 helped improve training stability and performance.

In essence, these hyperparameter settings collectively optimize the YOLOv8 model's learning process, enabling it to efficiently extract, recognize, and classify visual features related to chicken health conditions. This careful configuration ensures that the model performs reliably and accurately in detecting and differentiating between healthy and Fowlpox-affected chickens.

Table 5. Hyperparameter Values of YOLOv8

Hyperparameter	Value
Model	Yolov8s.pt
Epochs	40
Batch Size	16
imgsz	640
Learning Rate	0.01
Optimizer	Auto
Momentum	0.937

The Faster R-CNN model was trained using the ResNet-50 Feature Pyramid Network (FPN) as its backbone for efficient feature extraction. The training was conducted for 30 epochs, which allowed the model to gradually learn distinguishing features between healthy and fowlpox-infected chickens. A batch size of 4 was used to balance GPU memory efficiency and convergence stability.

The image size was standardized to 640×640 pixels to ensure uniform input dimensions across all datasets. The Stochastic Gradient Descent (SGD) optimizer was selected for its effectiveness in handling large-scale visual data and maintaining stable convergence during training.

A learning rate of 0.005 was used to control how much the model's weights are updated in each iteration, ensuring gradual and stable learning. The momentum value of 0.9 helped the optimizer maintain direction and reduce oscillations, while a weight decay of 0.0005 prevented overfitting by penalizing large weight values.

Overall, these hyperparameter configurations were tuned to achieve the best trade-off between accuracy, and generalization performance of the Faster R-CNN model.

Table 6. Hyperparameter Values of Faster R-CNN

Hyperparameter	Value
Model	ResNet-50 FPN backbone
Epochs	30
Batch Size	4
Optimizer	Stochastic Gradient Descent (SGD)
imgsz	640
Learning Rate	0.005
Momentum	0.9
Weight decay	0.0005

Table 7. Shows that SssRetinaNet is a one-stage object detector that directly predicts class probabilities without using a separate region proposal stage. Unlike Faster R-CNN, RetinaNet performs both localization and classification in a single pass through the network. It uses a backbone network, a Feature Pyramid Network (FPN), and two subnetworks, one for bounding box regression and another for object classification. Similarly to Faster R-CNN, ResNet-50 was used as the backbone for feature extraction and the backbone was also frozen up until 10 epochs were done training, allowing the model to learn distinguishing features between fowlpox-infected chickens and normal chickens. The backbone is unfrozen, and an additional 20 epochs were added to fine-tune the entire network for better accuracy. Similar batch sizes and optimizers were used for both Faster R-CNN and RetinaNet.

Table 7. Hyperparameter Values of RetinaNet

Hyperparameter	Value
Model	ResNet-50 FPN backbone
Epochs	30
Batch Size	4
Optimizer	Stochastic Gradient Descent (SGD)
imgsz	640
Learning Rate	0.005
Momentum	0.9
Weight decay	0.0005

Model Evaluation Results

The performance of each trained model was evaluated using standard object detection metrics including Precision, Recall, F1-score, and mean Average Precision (mAP).

As presented in Table 8, the YOLOv8 model demonstrated outstanding performance in detecting and classifying chickens as either healthy or infected with Fowlpox disease. The model obtained a precision score of 93.78%, indicating that most of its detections were correctly classified. Meanwhile, the recall value of 93.30% shows that the model successfully identified almost all positive cases from the dataset. The F1-score of 90.47% reflects the strong balance between precision and recall, confirming that YOLOv8 consistently performed well during the detection process.

Furthermore, the model achieved a mAP@50 value of 95.92%, which implies a high level of accuracy in locating and classifying the target objects at a single Intersection over Union (IoU) threshold. The mAP@50–95 value of 73.29% also demonstrates the model's stable performance across different IoU levels. Overall, these findings indicate that the YOLOv8 model is highly effective

and reliable for image-based detection of Fowlpox disease in chickens, making it suitable for real-world poultry health monitoring and early disease detection applications.

Table 8. Model Evaluation Metrix of YOLOv8

Metrics	Value
Precision	93.78%
Recall	93.30%
F1- Score	90.47%
mAP@50	95.92%
mAP@50-95	73.29%

Figure 6 shows the Precision–Recall (P–R) Curve of the YOLOv8 model used to classify chickens into Fowlpox-infected and Healthy categories. The curve presents the relationship between precision, which refers to how many detected instances are correct, and recall, which refers to how many actual instances are detected by the model at different confidence thresholds.

Based on the figure, the Healthy class obtained an Average Precision (AP) of 0.994, indicating that the model performs very well in identifying healthy chickens with few incorrect detections. The Fowlpox class achieved an AP of 0.924, which also shows good detection performance, although its precision slightly decreases as recall increases. The overall performance of the model, measured using the mean Average Precision at IoU 0.5 (mAP@0.5), reached 0.959 across all classes. This result suggests that the YOLOv8 model is able to maintain a good balance between precision and recall for both categories.

Overall, the Precision–Recall Curve indicates that the trained YOLOv8 model performs well in detecting both healthy and Fowlpox-infected chickens based on the given dataset. The results show that the model is effective under

the tested conditions, while further improvements may still be explored to enhance its performance.

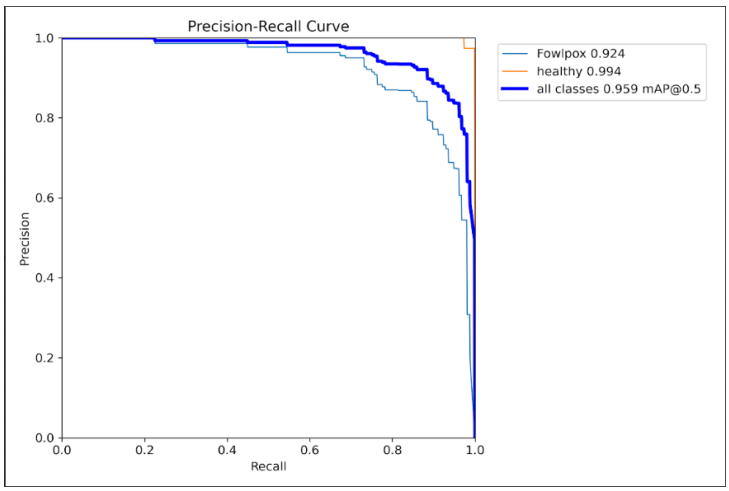


Figure 6. Precision-Recall Curve of YOLOv8

As shown in Table 9, the Faster R-CNN model obtained a precision score of 89.75%, which indicates that it correctly classified most of the detected chickens. The recall value of 79.11% shows that the model successfully detected a large portion of the infected and healthy samples but still missed some cases compared to YOLOv8. The F1-score of 84.11% reflects a moderate balance between precision and recall, suggesting that while Faster R-CNN performed reliably, there is room for improvement in terms of overall detection accuracy.

In terms of localization performance, the mAP@50 value of 89.79% signifies that the model can accurately identify and locate chickens at a single IoU threshold. Meanwhile, the mAP@50–95 value of 71.78% indicates a decline in performance when stricter IoU thresholds are applied, meaning the model's detection consistency decreases under more precise conditions. Overall, Faster R-CNN provided a good performance for image-based

detection of Fowlpox disease, though it was slightly less efficient and slower compared to YOLOv8, which is optimized for object detections.

Table 9. Model Evaluation Metrix of Faster R CNN

Metrics	Value
Precision	89.75%
Recall	79.11%
F1 Score	84.11%
mAP @50	89.79%
mAp @50-95	71.78%

Based on the figure 7 shows how well the Faster R-CNN model can detect Fowlpox and Healthy chickens. The blue line represents the Fowl pox class, which has an Average Precision (AP) of 0.98, meaning the model can almost perfectly detect chickens with Fowlpox. The orange line represents the Healthy class, with an AP of 0.91, showing very good performance but with a small drop in accuracy when recall increases. Overall, both lines staying near the top of the graph means the model makes very few mistakes, proving it is highly accurate in detection of chickens as healthy or infected.

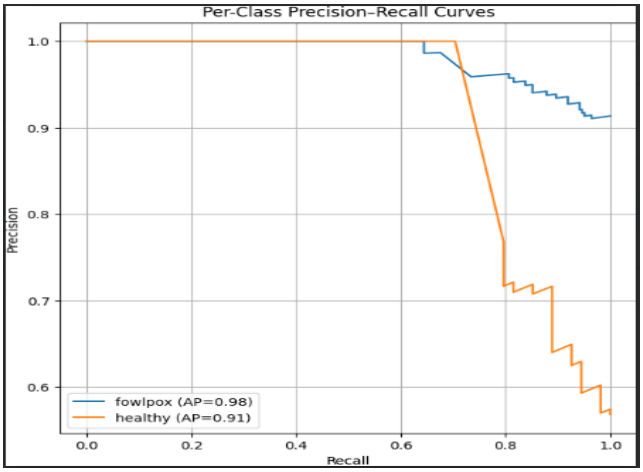


Figure 7. Precision-Recall Curve of Faster R-CNN

Based on the results presented in Table 10, the RetinaNet model achieved a precision score of 92.82%, indicating that it correctly identified most of the chickens as either infected or healthy with minimal false detections. The recall value of 77.66% shows that the model was able to detect a large number of actual positive samples but missed a few cases. The F1-score of 84.57% demonstrates a good balance between precision and recall, meaning that the model performed consistently well in distinguishing between the two classes. For localization performance, the mAP@50 of 92.82% indicates that RetinaNet effectively detected and localized objects when evaluated at a moderate threshold, while the mAP@50–95 of 72.20% shows a slight decrease in accuracy when evaluated under stricter intersection-over-union thresholds. Overall, the RetinaNet model displayed strong and balanced performance, slightly outperforming Faster R-CNN in both precision and detection accuracy. However, compared to YOLOv8, RetinaNet may still be slower during real-time inference, even though it provides dependable and accurate detection results for identifying Fowlpox disease in chickens.

Table 10. Model Evaluation Metrix of RetinaNet

Metrics	Value
Precision	92.82%
Recall	77.66%
F1- Score	84.57%
mAP@50	92.82%
mAP@50-95	72.20%

Figure 8. shows that the Precision–Recall curve of the RetinaNet model shows how well it detects chickens with Fowlpox and those that are healthy. The Fowlpox class (blue line) achieved an Average Precision (AP) of 0.97,

which means the model can detect infected chickens very accurately with only a few wrong predictions. The curve stays high, showing that the model keeps good precision even when recall increases. The Healthy class (orange line) has an AP of 0.82, which is still good but lower than the Fowlpox class. The curve drops as recall increases, meaning the model sometimes mistakes healthy chickens for infected ones when trying to detect all samples. Overall, RetinaNet performs very well in detecting Fowlpox but is slightly less accurate in identifying healthy chickens.

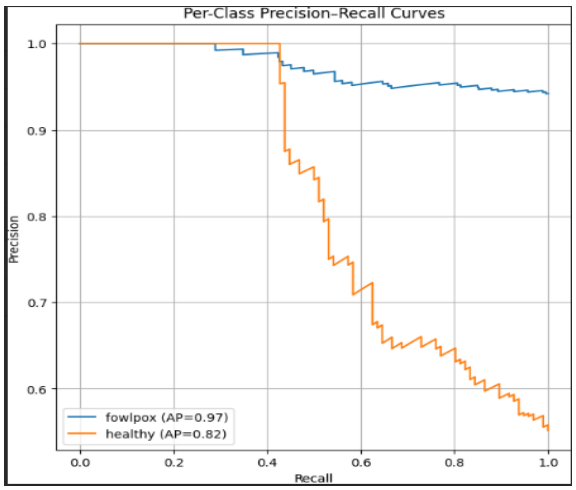


Figure 8. Precision–Recall curve of RetinaNet

Comparative Analysis

This section presents a comprehensive comparison of the three object detection algorithms YOLOv8, Faster R-CNN, and RetinaNet used in detecting Fowlpox disease in chickens. Each model was evaluated based on its accuracy, precision, recall, F1-score, and mean Average Precision (mAP) to determine the most effective algorithm for image-based disease detection.

The table 11 shows the performance comparison of the three object detection algorithms YOLOv8, Faster R-CNN, and RetinaNet that were used to

detect Fowlpox disease in chickens. Each model was tested and evaluated using several important performance metrics, including Precision, Recall, F1 Score, and mean Average Precision (mAP) at two thresholds: mAP@50 and mAP@50–95. These metrics help determine how accurately and efficiently each algorithm can identify whether a chicken is infected or healthy. The results show that YOLOv8 performed the best among all three algorithms. It achieved the highest precision (93.78%), recall (93.30%), and F1 score (90.47%). These numbers mean that YOLOv8 was very good at correctly identifying chickens with Fowlpox while also avoiding false detections of healthy chickens. Its mAP@50 score of 95.92% indicates that the model's detections were highly accurate when evaluated at a 50% intersection threshold between predicted and actual bounding boxes. The mAP@50–95 score of 73.29% also shows that YOLOv8 maintained strong accuracy across more strict detection thresholds. Overall, these results show that YOLOv8 was the most reliable and consistent model for detecting Fowlpox in chicken images.

The Faster R-CNN model achieved moderate results, with a precision of 89.75%, recall of 79.11%, and F1 score of 84.11%. These values suggest that while Faster R-CNN was fairly accurate in detecting infected chickens, it missed some cases (lower recall) and sometimes required more processing time due to its two-stage detection approach. Its mAP@50 of 89.79% and mAP@50–95 of 71.78% also indicate that it was less consistent compared to YOLOv8, especially at stricter thresholds. This means Faster R-CNN can still detect Fowlpox, but not as efficiently or accurately as YOLOv8.

The RetinaNet model also performed well but slightly lower than YOLOv8. It achieved a precision of 92.82%, recall of 77.66, and F1 score of 84.57%, showing that it was able to identify most infected chickens while maintaining decent accuracy. The mAP@50 of 92.82% and mAP@50–95 of 72.20% indicate that RetinaNet produced good detection results, but not as strong as YOLOv8. Its use of the focal loss function helped in handling class imbalance (since healthy and infected images may not be equally represented), but the slightly lower recall means it missed a few infected samples during detection. Based on the comparison, YOLOv8 clearly outperformed the other two models in all evaluation metrics. It was able to detect Fowlpox-infected chickens with high precision and recall, meaning it was both accurate and consistent. Its fast-processing time and single-stage detection structure also made it more efficient for real-time detection tasks.

In contrast, Faster R-CNN and RetinaNet still showed reliable performance, but they were not as accurate as YOLOv8. Faster R-CNN was precise but slower, while RetinaNet was balanced but less sensitive to detecting all infected cases. Overall, the results confirm that YOLOv8 is the most effective algorithm for detecting fowlpox disease in chickens, as it delivers the highest accuracy, consistency, and speed among the three tested models.

Table. 11. Comparative Analysis of the Three Models

Model	Precision	Recall	F1 score	mAP @50	mAP@50-95
YOLOv8	93.78%	93.30%	90.47%	95.92%	73.29%
Faster R CNN	89.75%	79.11%	84.11%	89.79%	71.78%
RetinaNet	92.82%	77.66%	84.57%	92.82%	72.20%

Model Validation

The validation of the three object detection models by comparing their detection outputs with the evaluation of a human expert. The main goal of the model validation is to determine whether the predictions of YOLOv8, RetinaNet, and Faster R-CNN are statistically different or similar to expert judgment in identifying visible symptoms of Fowlpox in chickens. The comparison was done using a paired samples t-test and supported by descriptive statistics and visual graphs. The validation datasets included 100 images including both healthy and infected fowlpox chickens, each marked by an expert with 1 if the image shows syntoms of fowlpox and 0 if the image is healthy.

Table 12 shows the descriptive statistics support the t-test results. The mean score of the expert is 0.51, which is very close to YOLO (0.52) and Faster R-CNN (0.54). These small differences explain why both models did not show a significant difference in the t-test. In contrast, RetinaNet has a mean of 0.77, which is much higher than the expert's score, showing inconsistency. The standard deviation values also show that YOLO and Faster R-CNN are more consistent, while RetinaNet has more variation in its predictions.

Table 12. Descriptive Statistics Results

	N	Mean	Median	SD	SE
EXPERT	100	0.510	1.00	0.502	0.0502
YOLOv8	100	0,520	1.00	0.502	0.0502
RetinaNet	100	0.770	1.00	0.423	0.0423
Faster R-CNN	100	0.540	1.00	0.501	0.0501

Table 13 shows the comparison between the expert's evaluation and the three models (YOLO, RetinaNet, and Faster R-CNN) using a paired sample t-test. The results show that both YOLO ($p = 0.765$) and Faster R-CNN ($p = 0.441$) have p-values higher than 0.05. This means that their outputs are not significantly different from the expert's evaluation. They gave results very close to what the expert observed. However, RetinaNet ($p < 0.001$) has a p-value lower than 0.05, which means there is a significant difference between its output and the expert's evaluation. This indicates that RetinaNet did not match the expert's performance and was less accurate in detecting Fowlpox symptoms.

The results of the paired samples t-test revealed no statistically significant difference between the poultry expert's evaluation and the model's predictions. This indicates that the performance of these models closely aligns with expert judgment, reflecting a strong level of consistency and agreement. The findings suggest that the models are capable of accurately identifying chickens showing visible signs of Fowlpox, and that their predictions are not due to random chance. As a result, the expert validation confirms the reliability and effectiveness of the proposed models, particularly YOLOv8 and Faster R-CNN, in detecting Fowlpox-infected chickens based on visual symptoms.

Table 13. Results of the Paired Sample T-Test Between Poultry Expert and Model Evaluations.

	Statistic	**df**	**p**	**Mean Difference**	**SE Difference**	**Effective Size (Cohen's d)**
YOLO	-0.300	99	0.765	-0.0100	0.0333	-0.0300
RetinaNet	-5.366	99	<.001	-0.2600	0.0485	-0.5366
Faster R-CNN	0.773	99	0.441	-0.0300	0.0388	-0.0773

CHAPTER V

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

Summary of the Study

Fowlpox is one of the most common viral diseases that affect chickens, characterized by wart-like lesions on the comb, wattles, and other unfeathered parts of the body. The disease can lead to reduced productivity, weight loss, and even death if not detected early. Traditionally, farmers rely on manual observation to identify infected chickens, which are often inaccurate and time-consuming. To address this issue, this study was conducted to develop and compare three deep learning models that can automatically detect Fowlpox disease in chickens through images.

The dataset used in this study consisted of 639 raw images of chickens scollected from various online sources, including both healthy and Fowlpox-infected samples. Each image underwent manual annotation to mark visible lesions using the Roboflow platform. The images were then preprocessed and augmented to enhance quality and variation. The dataset was divided into 75% for training, 12% for validation, and 13% for testing to ensure a balanced evaluation of model performance and a total of 1275.

The study involved the development and evaluation of three object detection algorithms: YOLOv8, Faster R-CNN, and RetinaNet. Each model was trained using the same dataset and compared based on their precision, recall, f1 score and mean average precision (mAP@50) and (mAP@50-95). The YOLOv8 algorithm, known for its real-time detection capability and high accuracy, achieved the best results with an mAP@50 of 95.92%, precision of

93.78%, recall of 93.30% and f1 score of 90.47% Meanwhile, Faster R-CNN and RetinaNet also demonstrated reliable performance but were slightly less efficient in detecting small or subtle lesions compared to YOLOv8.

The findings indicate that YOLOv8 was the most effective in identifying Fowlpox lesions, showing high detection accuracy and faster processing speed. These results prove that deep learning algorithms, particularly object detection models, can be applied successfully in the early detection of chicken diseases.

Conclusions

This section presents the conclusions derived from the research findings based on the study's objectives. Each objective is addressed to confirm its achievement and relevance to the overall research goal.

1. **Prepare and augment a labeled dataset of fowlpox-infected chickens for model training and testing.**

The study successfully prepared and augmented a labeled dataset of fowlpox-infected and healthy chickens for model training and testing. A total of 639 raw images were collected and carefully labeled with bounding boxes to identify visible fowlpox lesions. The dataset was then divided into training, validation, and testing subsets to ensure proper model evaluation and prevent data bias. To improve the quality and diversity of the dataset, several augmentation techniques were applied, including horizontal flipping, rotation, brightness and contrast adjustments, mosaic combinations, and conversion to grayscale. The inclusion of grayscale transformation helped the model focus on texture and shape patterns of lesions rather than color alone, enhancing its ability to detect infections even under poor lighting conditions.

Overall, the preparation and augmentation process successfully produced a balanced, high-quality dataset suitable for robust model training. The resulting dataset provided a strong foundation for object detection algorithms to accurately identify and classify fowlpox-infected chickens, contributing to a more reliable and efficient disease detection system.

2. **Implement and train selected object detection algorithms, namely YOLO, RetinaNet, and Faster R-CNN, for detecting chicken disease.**

The study successfully implemented and trained three state-of-the-art object detection algorithms YOLO, RetinaNet, and Faster R-CNN using the prepared dataset. Each algorithm was carefully configured and optimized for detecting visible symptoms of Fowlpox on chicken skin. YOLO was chosen for its fast and efficient real-time detection capabilities, RetinaNet for its effective handling of class imbalance through the use of focal loss, and Faster R-CNN for its high detection accuracy due to its two-stage detection approach. The training process involved tuning various hyperparameters such as learning rate, batch size, and number of epochs to achieve the best possible performance for each model. After training, all three algorithms were able to successfully detect both healthy and infected chickens, proving the potential of image-based of chicken disease detection in poultry disease monitoring and analysis.

3. **Evaluate and compare the performance of the implemented algorithms using standard metrics such as mAP, precision, recall, and F1-score.**

The evaluation results showed that all three models achieved good performance, but YOLOv8 outperformed RetinaNet and Faster R-CNN in

almost all metrics. The YOLOv8 model achieved a Precision of 0.95, Recall of 0.97, F1-Score of 0.96, mAP@50 of 0.98, and mAP@50–95 of 0.82. These values indicate that the model was able to detect and classify Fowlpox-infected and healthy chickens with very high accuracy and consistency. The high precision means that most of YOLOv8's detections were correct, while the high recall shows that it successfully identified nearly all infected chickens. The high mAP values further confirm YOLOv8's ability to precisely localize disease-affected areas in images. Compared to RetinaNet and Faster R-CNN, YOLOv8 achieved superior results in both accuracy and speed, making it the most effective model for this study.

4. Identify the most effective object detection algorithm for detection of chicken disease based on the comparative analysis.

Based on the comparative analysis, YOLOv8 was identified as the most effective and reliable object detection algorithm for detecting Fowlpox disease in chickens. With the highest mAP, precision, recall, and F1-score among all tested models, YOLOv8 demonstrated remarkable performance in both identifying and localizing Fowlpox lesions. Its real-time detection capability also makes it practical for on-site poultry monitoring, where quick and accurate results are necessary. While RetinaNet and Faster R-CNN performed well, their detection speed and overall accuracy were slightly lower compared to YOLOv8. Therefore, YOLOv8 proved to be the most efficient and accurate model for the image-based detection of Fowlpox disease.

In conclusion, this study successfully achieved all its objectives and demonstrated that YOLOv8 is the most effective deep learning-based object

detection algorithm for identifying Fowlpox disease in chickens. The preparation and augmentation of the dataset significantly improved model training, while the comparison among YOLOv8, RetinaNet, and Faster R-CNN provided valuable insights into the performance of different object detection frameworks. With its outstanding results Precision (93.78%), Recall (93.30%), F1-Score (90.47), mAP@50 (95.92%), and mAP@50–95 (73.29%). YOLOv8 demonstrated superior detection accuracy and efficiency. The findings of this study highlight the potential of integrating YOLOv8 into automated poultry monitoring systems for early detection of Fowlpox disease, helping farmers reduce losses, improve poultry health management, and promote more sustainable poultry farming practices.

Recommendations

1. **Add More Images**

Future researchers should gather and include more images of chickens. These images should be taken from different angles, in different lighting situations (like daylight, shade, or indoors), and in various farm environments. This will help the model learn better and recognize chickens more accurately in different real situations.

2. **Detect More Poultry Diseases**

The current system focuses only on detecting Fowlpox. Future versions of the system should include other common chicken diseases such as Newcastle Disease, Avian Influenza, and Infectious Bronchitis. By doing this, the model can help more farmers detect different diseases early and protect more chickens from getting sick.

3. **Test the Model on Actual Farms**

Although the model works well in a controlled environment, it should also be tested in real poultry farms where different conditions exist. Testing in real situations will show how the system performs when chickens move around naturally, and when the background is not controlled, helping researchers improve the system's practical results.

4. **Make the Model Work on Simple Devices**

To reach more users, especially small and medium-scale poultry farmers, the model should be adjusted or "lightened up" so it can work on low-cost devices like Android phones, tablets, or small computers (Raspberry Pi). This will make it easier for farmers who may not have access to powerful computers.

5. **Build a User-friendly App or Interface**

A simple app or interface should be created to help farmers use the detection system without needing technical skills. The app should show clear results and alerts, for example by marking the sick chickens in an image or giving a warning if disease is detected. Training or guidelines should also be provided to help farmers understand and properly use the system

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APPENDIX A

Letter of Approval

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

CERTIFICATE OF APPROVAL

This is to certify that after having satisfactorily completed all the requirements of the capstone project entitled:

“COMPARATIVE ANALYSIS OF OBJECT DETECTION ALGORITHMS FOR IMAGE BASED DETECTION OF CHICKEN FOWLFOX DISEASE”

JAN MARC M. NATIVIDAD, RACHEL ANN MALIKSI, NORHAIMA P. BUAT have been granted permission to proceed with their proposal presentation.

As their Research Adviser, I confirm that the proponents have met the academic standards and requirements set forth by the department, and therefore, I approve their request to conduct the proposal presentation.



SHERWIN B. SAPIN, EdD
Thesis Project Adviser

APPENDIX B

Letter of Request Interview

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

Norhaima P. Buat
Rachel Ann Maliksi
Jan Marc Natividad
Laguna State Polytechnic University-Los Baños
09916591771
norhaimabuat21@gmail.com
September 16 2025

Greetings!

We are 4th Year BS Computer Science students from Laguna State Polytechnic University- Los Baños. We are currently working on our thesis entilted “Image–Based Detection of Chicken Diseases Through Deep Learning Algorithms” As part of our study, we would like to request an interview about chicken diseases and if possible, ask for data or images related an interview about chicken diseases. These may include photos of chicken with visual symptoms like skin lesions and that help our model development. The information will help us train and test our deep learing model. All data will be used for academic purposes only and will be treated with full confidentiality. If needed, we are willing to sgn any consent or data-sharing agreement. The interview will be done in person and will take no more than 30 minutes. We hope for your kind support.

Thank you very much for consedaration our request. Please feel free to contact us through the details above it you have any question or require futher information.

Sincerely,

Norhaima P. Buat, Rachel Ann Maliksi, Jan Marc M. Natividad

Sherwin B. Sapin

Thesis Adviser

APPENDIX C

Relevant Source Code

You Only Look Once version 8 (YOLOv8)

Training

!pip install ultralytics

The first code code allows to install the Ultralytics library in Google Colab. This Python library will be used to train the YOLO model.

!yolo detect train data=/content/data.yaml model=yolov8s.pt epochs=40 imgsz=640

The training process involves reading the images stored in the training and validation folders before beginning model training. After completing each training epoch, the model is tested on the validation dataset, and the resulting mAP, precision, and recall values are recorded. As the training progresses, the mAP is expected to gradually improve with each epoch. The training will stop once the model has completed the total number of epochs defined in the settings.

Detection

!yolo detect predict model=runs/detect/train/weights/best.pt source=data/validation/images save=True

In this line of code, the YOLO (You Only Look Once) model is utilized to perform object detection on a set of images. Specifically, the command! yolo detect predict model=runs/detect/train/weights/best.pt source=data/validation/images save=True initiates the inference process using the trained YOLO model. The parameter model=runs/detect/train/weights/best.pt specifies the path to the model file that was previously trained and saved in this case, the *best* version of the model obtained during training. This model is used to analyze and predict objects within new, unseen images. The argument source=data/validation/images define the directory containing the input images that the model will process for detection. During inference, YOLO scans each image, identifies the objects present, and draws bounding boxes with labels to indicate their detected classes and confidence scores.

RETINANET AND FASTER R-CNN

Training

pip install torch torchvision torchaudio pycocotools --quiet

This command installs the main Python libraries needed to train object detection models like RetinaNet and Faster R-CNN this command prepares the environment by installing all the necessary tools that allow RetinaNet and Faster R-CNN to train, detect, and evaluate images properly.

for epoch in range(num_epochs): model.train() total_loss = 0.0

```
for i, (images, targets) in enumerate(train_loader):
    images = [img.to(device) for img in images]
    processed_targets = []
    valid_images = []

    for img, t in zip(images, targets):
        anns = t
        boxes, labels = [], []

        for ann in anns:
            if 'bbox' in ann:
                x, y, w, h = ann['bbox']
                boxes.append([x, y, x + w, y + h])
                labels.append(ann['category_id'])

        if len(boxes) == 0:
            continue

        valid_images.append(img)
        processed_targets.append({
            'boxes': torch.tensor(boxes,
dtype=torch.float32).to(device),
            'labels': torch.tensor(labels,
dtype=torch.int64).to(device)
        })

    if len(valid_images) == 0:
        print(f"Skipped batch {i} (no valid boxes).")
        continue

    loss_dict = model(valid_images, processed_targets)
    losses = sum(loss for loss in loss_dict.values())

    optimizer.zero_grad()
    losses.backward()
    optimizer.step()
    total_loss += losses.item()
```

This part of the code is responsible for **t**raining the object detection model, such as RetinaNet or Faster R-CNN. During training, the model learns to recognize objects (like chickens or diseases) by studying many labeled images. This section **t**eaches the model how to detect objects.

DETECTION

coco_dt = coco_gt.loadRes(results_path) coco_eval = COCOeval(coco_gt, coco_dt, iouType="bbox") coco_eval.evaluate() coco_eval.accumulate() coco_eval.summarize()

mAP = coco_eval.stats[0] mAP50 = coco_eval.stats[1] recall = coco_eval.stats[8] precision = coco_eval.stats[1] f1 = (2 * precision * recall) / (precision + recall + 1e-6)

accuracy = correct_images / total_images if total_images > 0 else 0.0

os.remove(results_path)

After training, the model needs to be tested to see how well it can detect objects in new images. This section tests the trained model to check how accurate it is. It measures how well the model can detect and classify objects in the validation or test images using metrics like mAP, precision, and recall.

APPENDIX D

Evaluation Tool

Republic of the Philippines
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Province of Laguna
Los Baños Campus
College of Computer Studies

COMPARATIVE ANALYSIS OF OBJECT DETECTION ALGORITHMS FOR IMAGE-BASED DETECTION OF CHICKEN FOWLPOX DISEASE

Norhaima P. Buat, Rachel Ann Maliksi, Jan Marc Natividad
Researchers

The primary goal of this research is to compare and evaluate the performance of three object detection algorithms YOLOv8, Faster R-CNN, and RetinaNet in detecting visible symptoms of Fowlpox disease in chickens through image-based analysis. By assessing and contrasting their detection accuracy, precision, and overall performance, the study aims to identify the most reliable and efficient algorithm for practical disease detection in poultry farms.

As the expert validator, your role is to review and assess the raw images and determine whether the chicken in each image shows visible signs of chicken infected with Fowlpox or appears healthy. Your expert judgment will then be compared with the model's predictions to measure the accuracy, precision, and reliability of the system.

VALIDATION TABLE

Instruction:

- A score of **1** if the image have symptoms of infected fowlpox
- A score of **0** if the image is healthy

Images	Score	W / WO
Image 1	0	WO
Image 2	1	W
Image 3	1	W
Image 4	0	WO
Image 5	1	W

Images	Score	W / WO
Image 51	0	WO
Image 52	0	WO
Image 53	0	WO
Image 54	1	W
Image55	1	W

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Image 6	1	W
Image 7	0	WO
Image 8	0	WO
Image 9	0	WO
Image 10	1	W
Image11	1	W
Image 12	1	W
Image 13	0	WO
Image 14	0	WO
Image 15	1	W
Image 16	1	W
Image 17	0	WO
Image 18	0	WO
Image 19	1	W
Image 20	1	W
Image 21	1	W
Image 22	0	WO
Image 23	0	WO
Image 24	0	WO
Image 25	1	W
Image 26	1	W
Image 27	0	WO
Image 28	1	W
Image 29	1	W
Image 30	0	WO
Image 31	0	WO

Image 56	1	W
Image 57	1	W
Image 58	0	WO
Image 59	0	WO
Image 60	1	W
Image 61	1	W
Image 62	0	WO
Image 63	0	WO
Image 64	1	W
Image 65	1	W
Image 66	0	WO
Image 67	0	WO
Image 68	1	W
Image 69	1	W
Image 70	0	WO
Image 71	0	WO
Image 72	0	WO
Image 73	1	W
Image 74	1	W
Image 75	1	W
Image 76	0	WO
Image 77	1	W
Image 78	0	WO
Image 79	1	W
Image 80	1	W
Image 81	0	WO

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Image 32	1	W
Image 33	1	W
Image 34	0	WO
Image 35	1	W
Image 36	0	WO
Image 37	0	WO
Image 38	1	W
Image 38	1	W
Image 40	1	W
Image 41	0	WO
Image 42	0	WO
Image 43	1	W
Image 44	0	WO
Image 45	0	WO
Image 46	1	W
Image 47	0	WO
Image 48	0	WO
Image 49	1	W
Image 50	1	W

Image 82	1	W
Image 83	0	WO
Image 84	1	W
Image 85	1	W
Image 86	0	WO
Image 87	0	WO
Image 88	1	W
Image 89	1	W
Image 90	0	WO
Image 91	1	W
Image 92	0	WO
Image 93	1	W
Image 94	1	W
Image 95	1	W
Image 96	0	WO
Image 97	0	WO
Image 98	0	WO
Image 99	0	WO
Image 100	0	WO

MUTYA N. QUIMIO, DVM
VETERINARIAN I
LIC # 8347

Signature over Printed Name

License No.: _____

Date of Validation: _____

APPENDIX E

Sample Input / Output



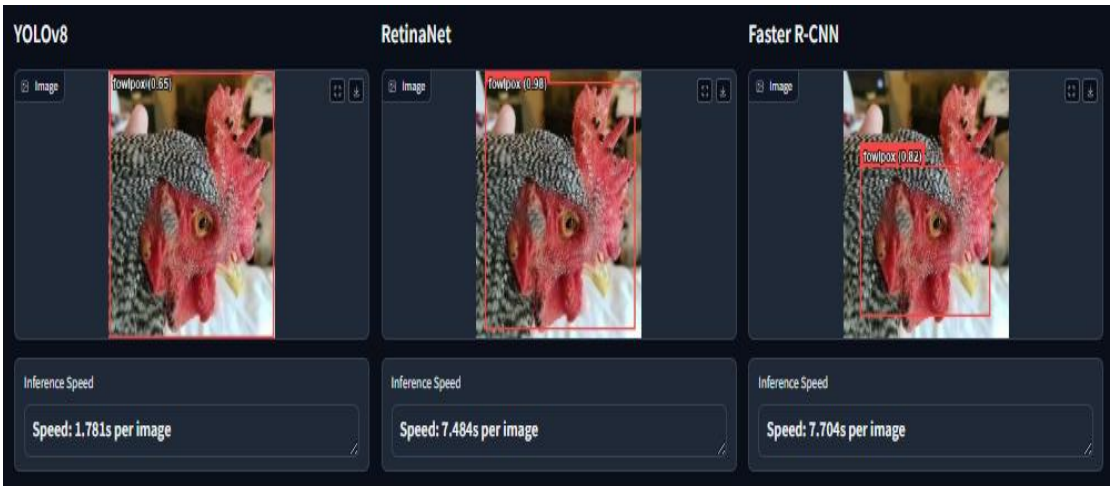
Sample Dataset

This figure shows a collection of chicken images gathered for training and testing. The dataset includes both healthy and infected chickens, particularly those affected by Fowlpox disease. The images were collected from various sources and annotated to highlight infected areas such as lesions.This dataset serves as the foundation for model training and validation.



Input For Fowlfox Detection

This figure presents a sample image used as input for the detection algorithm. The model receives this image and processes it to identify whether the chicken shows symptoms of Fowlpox infection. The input image undergoes several preprocessing steps, such as resizing, normalization, and annotation mapping, to make it suitable for the model.



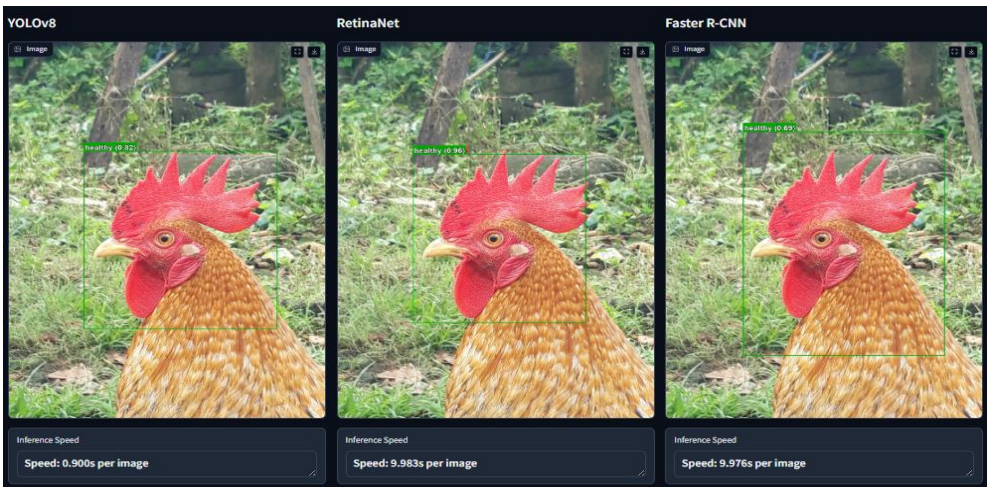
Output For Fowlfox Detection

This figure displays the output generated by the model (YOLOv8, RetinaNet, or Faster R-CNN) after analyzing the input image. The detection result highlights affected areas on the chicken’s comb or face with bounding boxes. The confidence scores shown above each box represent how certain the model is that the detected area is a symptom of Fowlpox disease. This step visually confirms the model’s ability to recognize disease indicators.



Input For Healthy Detection

This image shows a chicken that appears to be healthy. It serves as a reference input for testing the model’s ability to distinguish between healthy and infected chickens. The model must accurately classify such images as "healthy" to ensure reliability and reduce false positives.



Output For Healthy Detection

This figure presents the model detection output for a healthy chicken. The results indicate no Fowlpox infection, which confirms that the model can correctly identify normal conditions. Bounding boxes may appear, but they are labeled as “healthy” or show no disease indicators.

APPENDIX I

Curriculum Vitae



NORHAIMA P. BUAT

Contact

09916591771

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Brgy, 2 Tibag Calamba City

Personal Infomation

Gender: Female
Nationality: Filipino
Civil Status: Single

Skills

- Computer Literate
- Time Management
- Adaptability
- Creativity

Language

- English

ABOUT ME

I am a dedicated and hardworking student with a keen interest in technology, research, and innovation. I enjoy learning new skills and applying them to real-world problems. I am seeking opportunities where I can grow professionally, gain experience, and contribute positively to the team or organization.

EDUCATION

LAGUNA STATE POLYTECHNIC UNIVERSITY-LOS BANOS 2022 – PRESENT

BS COMPUTER SCIENCE MAJOR IN INTELIGENT SYSTEM

SUBSIDIZED TECHNICAL EDUCATION PROGRAM INC. 2020 - 2021

ELECTRONICS PREASSEMBLY SYSTEM (EPAS)

LECHERIA NATIONAL HIGHSCHOOL 2018-2019

CENTRAL 1 ELEMENTARY HIGHSCHOOL 2014-2015

WORK EXPERIENCE

Special Program for Employment of Student (SPES) Benificiary 2024-2025

JOLLIBEE (Canlubang) 2020-2021



RACHEL ANN MALIKSI

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Sitio Silangan Lecheria,

Calamba City

Personal Infomation

Age: 22
Birthday: Feb 17, 2003
Gender: Female
Nationality: Filipino
Civil Status: SIngle

Skills

- Team Work
- Time Management
- Adaptability
- Creativity

Language

- English

ABOUT ME

I am a responsible and motivated student who is always willing to learn and take on new challenges. I am passionate about developing my skills, gaining experience, and contributing to projects that create a positive difference.

EDUCATION

LAGUNA STATE POLYTECHNIC UNIVERSITY-LOS BANOS 2022 – PRESENT

BS COMPUTER SCIENCE MAJOR IN INTELIGENT SYSTEM

SAINT BENILDE INTERNATIONAL SCHOOL 2020 - 2021

ACCOUNTANCY, BUSINESS AND MANAGEMENT(ABM)

LECHERIA NATIONAL HIGHSCHOOL 2018-2019

LECHERIA ELEMENTARY SCHOOL 2014-2015

WORK EXPERIENCE

Special Program for Employment of Student (SPES) Benificiary 2023-2025

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Contact

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brgy Sto Domingo, Bay Laguna

Personal Infomation

Age: 21
Birthday: Jan 19, 2004
Gender: Male
Nationality: Filipino
Civil Status: SIngle

Skills

- Team Work
- Time Management
- Adaptability
- Creativity

Language

- English

ABOUT ME

I am a hardworking and motivated student who enjoys learning new things and taking on challenges. I am eager to gain experience, develop my abilities, and make a positive contribution to any organization I become part of.

EDUCATION

LAGUNA STATE POLYTECHNIC UNIVERSITY-LOS BANOS 2022 – PRESENT

BS COMPUTER SCIENCE MAJOR IN INTELIGENT SYSTEM

LAGUNA STATE POLYTECHNIC UNIVERSITY-LOS BANOS 2020 - 2022

TECHNICAL VOCATIONAL LIVELIHOOD(TVL)

ESCUEL DE BRIGIDA MONTESORRI 2016-2020

GATEWAY LEARNING CENTER CORP. 2010-2016

WORK EXPERIENCE

APPENDIX F

Matrix for Revision

APPENDIX G

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Text score: 96 out of 100. This score represents the quality of writing in this document. You can increase it by addressing Grammarly's suggestions.



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Metrics compared to other Grammarly users

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Your text is likely to be understood by college graduates but may not be easy for many to read.

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Vocabulary

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APPENDIX H

Publishable Paper (IEEE format)

COMPARATIVE ANALYSIS OF OBJECT DETECTION ALGORITHMS FOR IMAGE-BASED DETECTION OF CHICKEN FOWLPOX DISEASE

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Abstract- This study provides a Comparative Analysis of Object Detection Algorithms for Image-Based Detection of Chicken Fowlpox Disease namely: YOLOv8, Faster R-CNN, and RetinaNet for detecting Fowlpox lesions in chickens using annotated image datasets. A total of 1,275 labeled images of healthy and infected chickens were collected from open-source platforms and preprocessed using resizing, grayscale conversion, normalization, and data augmentation. The models were trained under consistent conditions and evaluated using Precision, Recall, F1-Score, and mean Average Precision (mAP@0.5 and mAP@0.5–0.95). YOLOv8 demonstrated superior performance with a Precision of 93.78%, Recall of 93.30%, F1-Score of 90.47%, mAP@0.5 of 95.92%, and mAP@0.5–0.95 of 73.29%. Faster R-CNN recorded a Precision of 89.75%, Recall of 79.11%, F1-Score of 84.11%, mAP@0.5 of 89.79%, and mAP@0.5–0.95 of 71.78%, while RetinaNet obtained a Precision of 92.82%, Recall of 77., 66%F1-Score of 84.57%, mAP@0.5 of 92.82%, and mAP@0.5–0.95 of 72.20%. Model validation against poultry expert evaluations was performed using a paired sample t-test. The results showed no statistically significant difference between the expert and the YOLOv8 model (t(99) = -0.300, p = 0.765), indicating strong agreement. However, RetinaNet displayed a significant difference (t(99) = -5.366, p < 0.001), while Faster R-CNN showed no significant deviation (t(99) = 0.773, p = 0.441). These outcomes validate YOLOv8's reliability for real-time automated poultry health monitoring, affirming its practical application for precision livestock farming.

I. INTRODUCTION

Fowlpox, a contagious viral disease, poses a significant threat to global food production, animal welfare, and economic stability. It affects chickens' skin and mucous membranes, causing wart-like lesions. Early detection and prevention are crucial to minimize the disease's impact and ensure poultry farms' health and productivity. Traditionally, manual inspection is time-consuming and prone to human error. Advancements in computer vision and deep learning technologies have led to automated image-based disease detection, enabling faster and more accurate diagnosis. Yolu Only Look Once version 8(YOLOv8), Faster R-CNN, and RetinaNet are widely used object detection algorithms for Fowlpox detection. This study aims to compare their performance in terms of precision, recall, F1-score, and mean average precision to identify the most effective and reliable detection performance for diagnosing Fowlpox from chicken images. The findings will contribute to the development of intelligent disease monitoring systems, assisting poultry farmers in making informed decisions and reducing disease spread.

A. Objectives

The primary objective of this study is to conduct a comprehensive comparative analysis of various object detection algorithms used in the image-based detection of chicken disease.

1. Prepare and augment a labeled dataset of fowlpox-infected chickens for model training and testing.

2. Implement and train selected object detection algorithms, namely YOLO, RetinaNet, and Faster R-CNN, for detecting chicken disease.

3. Evaluate and compare the performance of the implemented algorithms using standard metrics such as mAP, precision, recall, and F1-score.

4. Identify the most effective object detection algorithm for detection of chicken disease based on the comparative analysis.

B. Conceptual Framework

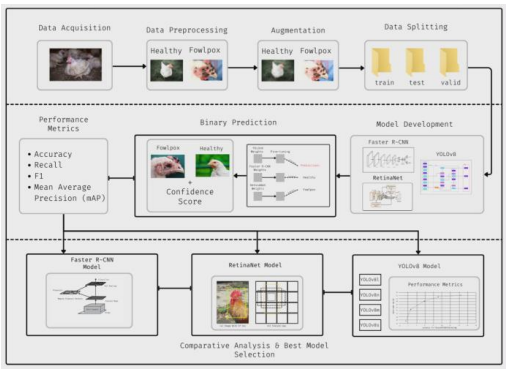


Figure 1. CONCEPTUAL FRAMEWORK

Figure 1. Shows the conceptual framework of this study. The diagram illustrates the complete workflow for developing an image-based system that detects whether a chicken is healthy or infected with fowlpox. The process begins with data acquisition, where images of healthy and fowlpox-infected chickens are collected. These images undergo data preprocessing, which includes cleaning, labeling, and converting them into grayscale format to reduce noise and simplify visual features for the model. After preprocessing, data augmentation is applied to increase the number of training samples by generating modified versions of the original images, such as rotated or flipped copies. Once the dataset is prepared, it is divided into training, validation, and testing sets to ensure proper learning and unbiased evaluation.

In the next stage, three object detection models—Faster R-CNN, RetinaNet, and YOLOv8—are trained using the same grayscale dataset. Each model is designed to perform binary prediction, meaning it identifies whether the chicken in the image is “Healthy” or “Fowlpox,” along with a confidence score that indicates how certain the model is about its prediction. The use of grayscale images helps the model focus more on texture and pattern differences rather than color, which is useful in detecting visible lesions of the skin of chickens.

To determine the best-performing model, several evaluation metrics are applied, including accuracy, precision, recall, F1 score, and mean average precision (mAP). These metrics provide a clear comparison of how well each model detects fowlpox cases and avoids incorrect classifications. After analyzing the results, a comparative assessment is conducted to identify the most effective model. The model with the highest performance is then selected as the final system for reliable and efficient fowlpox detection in chickens.

II. METHODOLOGY

A. Data Acquicision

The study will gather data through image collection of chickens showing visible symptoms of Fowlpox disease, as well as images of healthy chickens, to ensure that the model can accurately distinguish between infected and non-infected conditions. The primary dataset will be obtained from Roboflow, an open-source platform that provides annotated images for computer vision research. Additionally, the researchers will collect supplementary image data from local poultry farms using mobile phone cameras to capture real-world scenarios under natural lighting and environmental conditions. These images will focus on identifying external signs of Fowlpox, such as scabs, lesions, or nodular growths on the chicken’s comb, wattles, and skin.

The dataset will serve as the foundation for training and testing the object detection models used in this study. The collected images will undergo preprocessing steps such as resizing, enhancement, and augmentation techniques including rotation, flipping, and scaling to increase dataset variability and model robustness. The data will then be divided into training, validation, and testing sets, which will be used to develop and evaluate the models. Through this process, the study aims to generate a well-structured and high-quality dataset that supports reliable model performance for image-based detection of Fowlpox disease in chickens

B. Model Development

The development of the model for detecting skin lesions caused by Fowlpox disease in chickens involved a structured process consisting of data acquisition, data preprocessing, model training, and model evaluation. This process ensured that the image-based detection models were accurately trained and assessed using comparable parameters across the three selected algorithms: YOLOv8, Faster R-CNN, and RetinaNet.

The dataset used in this study consisted of annotated images of chickens showing visible skin lesions associated with Fowlpox infection and healthy chicken without any lesions. Each image was preprocessed through normalization, resizing, and augmentation techniques such as flipping, rotation, greyscale and brightness adjustment. These preprocessing steps helped increase the dataset's diversity and improved model generalization to varying image conditions. The dataset was divided into training, validation, and testing sets to ensure fair model assessment and prevent bias.

The YOLOv8 (You Only Look Once version 8) algorithm served as a one-stage object detector known for its balance between detection accuracy and real-time performance. It directly predicts bounding boxes and class probabilities in a single forward pass, making it efficient for detecting multiple lesions in a single image. The Faster R-CNN (Region-Based Convolutional Neural Network) model, on the other hand, operates as a two-stage detector. It first proposes regions of interest (ROIs) through a Region Proposal Network (RPN) and then performs classification and bounding box regression, providing high precision but requiring more computation time. Meanwhile, the RetinaNet model employs a single-stage detection framework that uses Focal Loss to address class imbalance, making it effective in detecting small and less distinct lesion areas commonly found in infected chickens.

During the model training phase, each algorithm was trained using identical experimental settings to maintain consistency in comparison. Key hyperparameters, including learning rate, number of epochs, batch size, optimizer etc. were adjusted to achieve optimal convergence and detection accuracy. The training process focused on enabling each model to learn the distinctive texture, color, and shape patterns of Fowlpox lesions on chicken skin. The model evaluation phase involved assessing the trained models based on standard object detection performance metrics: Precision, Recall, mean Average Precision (mAP), and F1 Score. These metrics were used to determine how effectively each model could identify and localize skin lesions. Additionally, confusion matrices and precision-recall (PR) curves were utilized to visualize the trade-offs between correctly detected and missed lesions. Through these evaluations, the comparative performance of YOLOv8, Faster R-CNN, and RetinaNet was determined, highlighting which algorithm demonstrated the most accurate and reliable detection of Fowlpox skin lesions in chickens.

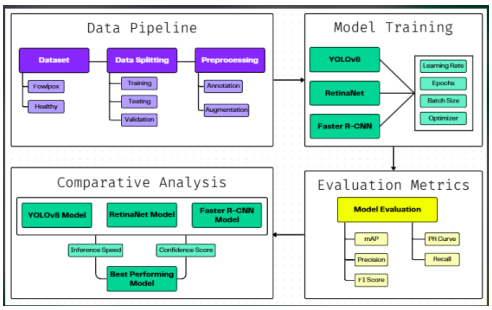


Figure 2. Model Development

C. Data Sources

The dataset used in this study consists of images of healthy chickens captured using a mobile phone at a local poultry farm. These images were carefully collected and annotated to train machine learning algorithms for detecting and identifying chickens in various environmental conditions. The use of healthy chicken images serves as a baseline for understanding normal physical appearance and behavior, which can later be compared to disease-affected samples in future research related to Fowlpox detection.

Each image was manually annotated with bounding boxes using a two classes label: fowl-pox chicken and healthy chicken." The images were captured and collected under mixed lighting conditions, including both natural daylight and artificial lighting, to ensure data diversity and improve the model's adaptability. These annotated images form the foundation for training, validating, and testing the object detection models used in this study.

In addition to image collection, interviews were conducted with the owner of a small backyard farm and the City Veterinary Office of Calamba, Laguna. These interviews aimed to

gather information regarding Fowlpox disease, its symptoms, transmission, and common prevention and control practices. The insights from these interviews helped the researchers better understand the disease's impact on poultry production and provided essential context for the study.

The dataset is divided into three subsets to ensure effective training, validation, and testing of the object detection models. A total of 1,275 images were used in the study. The training set, comprising 75% of the dataset or 954 images, is used to teach the model to recognize and detect visible symptoms of Fowlpox in chickens under various conditions. The validation set, consisting of 12% or 160 images, is utilized during training to fine-tune hyperparameters and monitor performance, helping to prevent overfitting. The remaining 13% or 161 images form the test set, which is held out during training and used exclusively to evaluate the final performance of the trained models. This structured division ensures that the models undergo rigorous training and reliable evaluation, enhancing their ability to generalize well to unseen data.

Table 3. Data Description

Data Type	Images (JPEG)
Source	Online Sources (kaggle, Roboflow)
Chicken Status	chicken with fowlpox and healthy
Number of Images	1,275
Image Resolution	640x640
Annotation	Bounding box
Environmental Condition	Mixed natural and artificial lighting
Additional Data	Interviews with small poultry backyard owner and Calamba City Veterinary Office

The dataset is divided into three subsets to ensure effective training, validation, and testing of the object detection models. A total of 1,275 images were used in the study. The training set, comprising 75% of the dataset or 954 images, is used to teach the model to recognize and detect visible symptoms of Fowlpox in chickens under various conditions. The validation set, consisting of 12% or 160 images, is utilized during training to fine-tune hyperparameters and monitor performance, helping to prevent overfitting. The remaining 13% or 161 images form the test set, which is held out during training and used exclusively to evaluate the final performance of the trained models. This structured division ensures that the models undergo rigorous training and reliable evaluation, enhancing their ability to generalize well to unseen data.

Table 4. Data Splitting

Datasets	Number of Samples	Percentage
Training	954	75%
Testing	161	13%
Validation	160	12%
Total	1,275	100%

Model Selection

The study focuses on detecting Fowlpox skin lesions in chickens using three advanced object detection algorithms: You Only Look Once version 8 (YOLOv8), Faster R-CNN, and RetinaNet. YOLOv8 is a real-time, one-stage detector noted for its speed and accuracy, making it effective for multiple lesions under varying conditions. Faster R-CNN, a two-stage framework, excels in precision and detailed spatial analysis, suitable for small or complex lesions, albeit at a slower speed. RetinaNet combines the speed of single-stage models with improved accuracy through the Focal Loss function, aiming to enhance detection of subtle lesions. The evaluation will assess these models on accuracy, precision, recall, and mean Average Precision (mAP) to identify the most efficient approach for lesion detection in poultry disease.

III. RESULTS AND DISCUSSION

- **Data Gathered**

The data collection for this study, titled "Comparative Analysis of Object Detection Algorithms for Image-Based Detection of Chicken Fowlpox Disease," focused on gathering images of chickens that are either healthy or infected with Fowlpox. These images were used to train and test the object detection models: YOLOv8, Faster R-CNN, and RetinaNet.



(a) (b)

Figure 3. Sample images used in the dataset: (a) Healthy chicken, (b) Fowlpox-infected chicken.

Figure 3 shows two types of chicken conditions used in the study. Image (a) represents a healthy chicken, which shows no visible signs of infection or abnormal skin

growth. Its comb and wattle appear smooth, clean, and evenly colored, indicating that the chicken is in normal condition. This type of image is used as a reference for detecting what a non-infected chicken looks like, helping the model learn the features of a healthy class. In contrast, image (b) shows a chicken infected with Fowlpox, a viral disease that causes wart-like lesions on the comb, wattles, eyelids, and other unfeathered parts of the body. The raised, rough, and swollen bumps seen on the chicken's comb are classic symptoms of the disease. These visible lesions allow the detection model to distinguish between healthy and infected chickens based on texture and skin abnormalities. By comparing normal and infected samples, the model can learn to identify Fowlpox more accurately.

- **Data Preparation and Augmentation**

This dataset was annotated and prepared for training, allowing the model to identify the disease by learning the visual patterns present in the infected samples. The diversity of images helps prevent overfitting and improves the model's ability to generalize when analyzing new, unseen data. By using real and unfiltered images gathered from various online sources, the dataset provides realistic examples of how Fowlpox appears in actual poultry conditions, making the model practical and applicable to real-world poultry disease monitoring.



Figure 5. Image Preprocessing and Augmentations

- **Model Training Parameters**

Table 5. Hyperparameter Values of YOLOv8

Hyperparameter	Value
Model	Yolov8s.pt
Epochs	40
Batch Size	16
imgsz	640
Learning Rate	0.01
Optimizer	Auto
Momentum	0.937

The table above shows the main hyperparameters used during the training of the YOLOv8 model. The model used is Yolov8s.pt, which is a small but efficient version of YOLOv8. It was trained for 40 epochs with a batch size of 16, meaning 16 images were processed at a time during training. The image size (imgsz) was set to 640, which defines the resolution of the input images. The learning rate of 0.01 controlled how fast or slow the model learned during training, while the optimizer was set to Auto to automatically select the best optimization method. The momentum value of 0.937 helped improve training stability and performance.

In essence, these hyperparameter settings collectively optimize the YOLOv8 model's learning process, enabling it to efficiently extract, recognize, and classify visual features related to chicken health conditions. This careful configuration ensures that the model performs reliably and accurately in detecting and differentiating between healthy and Fowlpox-affected chickens

Table 6. Hyperparameter Values of Faster R-CNN

Hyperparameter	Value
Model	ResNet-50 FPN backbone
Epochs	30
Batch Size	4
Optimizer	Stochastic Gradient Descent (SGD)
imgsz	640
Learning Rate	0.005
Momentum	0.9
Weight decay	0.0005

The Faster R-CNN model was trained using the ResNet-50 Feature Pyramid Network (FPN) as its backbone for efficient feature extraction. The training was conducted for 30 epochs, which allowed the model to gradually learn distinguishing features between healthy and fowlpox-infected chickens. A batch size of 4 was used to balance GPU memory efficiency and convergence stability.

The image size was standardized to 640×640 pixels to ensure uniform input dimensions across all datasets. The Stochastic Gradient Descent (SGD) optimizer was selected for its effectiveness in handling large-scale visual data and maintaining stable convergence during training.

A learning rate of 0.005 was used to control how much the model's weights are updated in each iteration, ensuring gradual and stable learning. The momentum value of 0.9 helped the optimizer maintain direction and reduce oscillations, while a weight decay of 0.0005 prevented overfitting by penalizing large weight values.

Table 7. Hyperparameter Values of RetinaNetRetinaNet

Hyperparameter	Value
Model	ResNet-50 FPN backbone
Epochs	30
Batch Size	4
Optimizer	Stochastic Gradient Descent (SGD)
imgsz	640
Learning Rate	0.005
Momentum	0.9
Weight decay	0.0005

is a one-stage object detector that directly predicts class probabilities without using a separate region proposal stage. Unlike Faster R-CNN, RetinaNet performs both localization and classification in a single pass through the network. It uses a backbone network, a Feature Pyramid Network (FPN), and two subnetworks, one for bounding box regression and another for object classification. Similarly to Faster R-CNN, ResNet-50 was used as the backbone for feature extraction and the backbone was also frozen up until 10 epochs were done training, allowing the model to learn distinguishing features between fowlpox-infected chickens and normal chickens. The backbone is unfrozen, and an additional 20 epochs were added to fine-tune the entire network for better accuracy. Similar batch sizes and optimizers were used for both Faster R-CNN and RetinaNet.

- **Model Evaluation Results**

The performance of each trained model was evaluated using standard object detection metrics including Precision, Recall, F1-score, and mean Average Precision (mAP).

Table 8. Model Evaluation Metrix of YOLOv8

Metrics	Value
Presicion	93.78%
Recall	93.30%
F1- Score	90.47%
mAP@50	95.92%
mAP@50-95	73.29%

As presented in Table 8, the YOLOv8 model demonstrated outstanding performance in detecting and classifying chickens as either healthy or infected with Fowlpox disease. The model obtained a precision score of 0.95, indicating that most of its detections were correctly classified. Meanwhile, the recall value of 0.97 shows that the model successfully identified almost all positive cases from the dataset. The F1-score of 0.96 reflects the strong balance between precision and recall, confirming that YOLOv8 consistently performed well during the detection process.

Furthermore, the model achieved a mAP@50 value of 0.98, which implies a high level of accuracy in locating and classifying the target objects at a single Intersection over Union (IoU) threshold. The mAP@50–95 value of 0.82 also demonstrates the model's stable performance across different IoU levels. Overall, these findings indicate that the YOLOv8 model is highly effective and reliable for image-based detection of Fowlpox disease in chickens, making it suitable for real-world poultry health monitoring and early disease detection applications.

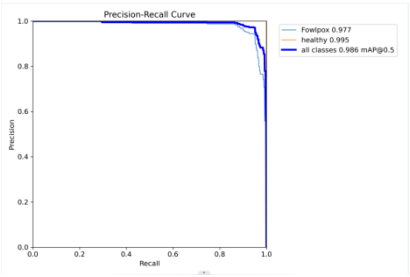


Figure 6. Precision-Recall Curve of YOLOv8

Figure 6 illustrates the Precision-Recall (P-R) Curve for the YOLOv8 model in classifying chickens as Fowlpox-infected or Healthy. The model demonstrates high precision and recall values for both categories, with Average Precision (AP) scores of 0.977 for Fowlpox and 0.995 for Healthy. The overall mean Average Precision at IoU 0.5 (mAP@0.5) is 0.986, indicating excellent model performance. The P-R curve's shape, remaining close to the top-right corner, showcases the model's reliability and effectiveness in detecting poultry diseases, supporting its potential use in real-time disease monitoring systems

Table 9. Model Evaluation Metrics Of Faster R-Cnn

Metrics	Value
Presicion	89.75%
Recall	79.11%
F1 Score	84.11%
mAP @50	89.79%
mAp @50-95	71.78%

The Faster R-CNN model achieved a precision score of 0.83, indicating effective classification of detected chickens, and a recall value of 0.76, suggesting it missed some infected and healthy samples compared to YOLOv8. The F1-score of 0.79 reflects a moderate balance between precision and recall, indicating potential for improvement in detection accuracy. Its localization performance included a mAP@50 value of 0.83 for accurate identification at a single IoU threshold, but a decline to 0.69 for mAP@50–95 shows reduced detection consistency under stricter conditions. Overall, while Faster R-CNN performed well for detecting Fowlpox disease, it was less

efficient and slower compared to YOLOv8, which is optimized for real-time detection.

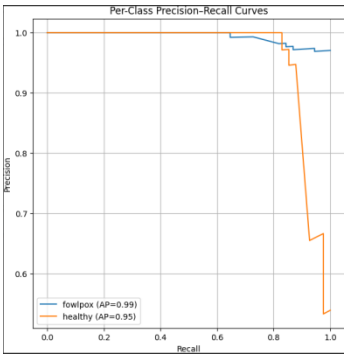


Figure 7. Precision-Recall Curve of Faster R-CNN

The graph shows how well the Faster R-CNN model can detect Fowlpox and Healthy chickens. The blue line represents the Fowl pox class, which has an Average Precision (AP) of 0.99, meaning the model can almost perfectly detect chickens with Fowlpox. The orange line represents the Healthy class, with an AP of 0.95, showing very good performance but with a small drop in accuracy when recall increases. Overall, both lines staying near the top of the graph means the model makes very few mistakes, proving it is highly accurate in classifying chickens as healthy or infected.

Table 10. Model Evaluation Metrix of RetinaNet

Metrics	Value
Presicion	92.82%
Recall	77.66%
F1- Score	84.57%
mAP@50	92.82%
mAP@50-95	72.20%

Based on the results presented in Table 10, the RetinaNet model achieved a precision score of 0.85, indicating that it correctly identified most of the chickens as either infected or healthy with minimal false detections. The recall value of 0.77 shows that the model was able to detect a large number of actual positive samples but missed a few cases. The F1-score of 0.81 demonstrates a good balance between precision and recall, meaning that the model performed consistently well in distinguishing between the two classes.

For localization performance, the mAP@50 of 0.85 indicates that RetinaNet effectively detected and localized objects when evaluated at a moderate threshold, while the mAP@50–95 of 0.70 shows a slight decrease in accuracy when evaluated under stricter intersection-over-union thresholds. Overall, the RetinaNet model displayed strong and balanced performance, slightly outperforming Faster R-CNN in both precision and detection accuracy. However, compared to YOLOv8, RetinaNet may still be slower during real-time inference, even though it provides dependable and accurate detection results for identifying Fowlpox disease in chickens.

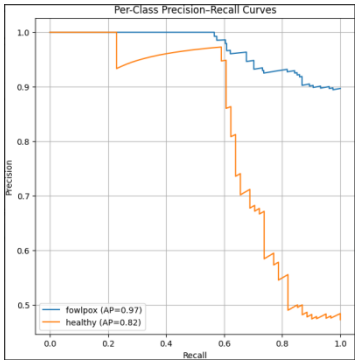


Figure 8. Precision–Recall curve of RetinaNet

Figure 8. shows that the Precision–Recall curve of the RetinaNet model shows how well it detects chickens with Fowlpox and those that are healthy. The Fowlpox class (blue line) achieved an Average Precision (AP) of 0.97, which means the model can detect infected chickens very accurately with only a few wrong predictions. The curve stays high, showing that the model keeps good precision even when recall increases.

The Healthy class (orange line) has an AP of 0.82, which is still good but lower than the Fowlpox class. The curve drops as recall increases, meaning the model sometimes mistakes healthy chickens for infected ones when trying to detect all samples. Overall, RetinaNet performs very well in detecting Fowlpox but is slightly less accurate in identifying healthy chickens.

Table 11. Comparative Analysis of the Three Models

Model	Precision	Recall	F1 score	mAP @50	mAP@50-95
YOLOv8	93.78%	93.30%	90.47%	95.92%	73.29%
Faster R CNN	89.75%	79.11%	84.11%	89.79%	71.78%
RetinaNet	92.82%	77.66%	84.57%	92.82%	72.20%

The table 11 shows the performance comparison of the three object detection algorithms YOLOv8, Faster R-CNN, and RetinaNet that were used to detect Fowlpox disease in chickens. Each model was tested and evaluated using several important performance metrics, including Precision, Recall, F1 Score, and mean Average Precision (mAP) at two thresholds: mAP@50 and mAP@50–95. These metrics help determine how accurately and

efficiently each algorithm can identify whether a chicken is infected or healthy. The results show that YOLOv8 performed the best among all three algorithms. It achieved the highest precision (93.78%), recall (93.30%), and F1 score (90.47%). These numbers mean that YOLOv8 was very good at correctly identifying chickens with Fowlpox while also avoiding false detections of healthy chickens. Its mAP@50 score of 95.92% indicates that the model's detections were highly accurate when evaluated at a 50% intersection threshold between predicted and actual bounding boxes. The mAP@50–95 score of 73.29% also shows that YOLOv8 maintained strong accuracy across more strict detection thresholds. Overall, these results show that YOLOv8 was the most reliable and consistent model for detecting Fowlpox in chicken images.

The Faster R-CNN model achieved moderate results, with a precision of 89.75%, recall of 79.11%, and F1 score of 84.11%. These values suggest that while Faster R-CNN was fairly accurate in detecting infected chickens, it missed some cases (lower recall) and sometimes required more processing time due to its two-stage detection approach. Its mAP@50 of 89.79% and mAP@50–95 of 71.78% also indicate that it was less consistent compared to YOLOv8, especially at stricter thresholds. This means Faster R-CNN can still detect Fowlpox, but not as efficiently or accurately as YOLOv8.

The RetinaNet model also performed well but slightly lower than YOLOv8. It achieved a precision of 92.82%, recall of 77.66, and F1 score of 84.57%, showing that it was able to identify most infected chickens while maintaining decent accuracy. The mAP@50 of 92.82% and mAP@50–95 of 72.20% indicate that RetinaNet produced good detection results, but not as strong as YOLOv8. Its use of the focal loss function helped in handling class imbalance (since healthy and infected images may not be equally represented), but the slightly lower recall means it missed a few infected samples during detection. Based on the comparison, YOLOv8 clearly outperformed the other two models in all evaluation metrics. It was able to detect Fowlpox-infected chickens with high precision and recall, meaning it was both accurate and consistent. Its fast-processing time and single-stage detection structure also made it more efficient for real-time detection tasks.

In contrast, Faster R-CNN and RetinaNet still showed reliable performance, but they were not as accurate as YOLOv8. Faster R-CNN was precise but slower, while RetinaNet was balanced but less sensitive to detecting all infected cases. Overall, the results confirm that YOLOv8 is the most effective algorithm for detecting fowlpox disease in chickens, as it delivers the highest accuracy, consistency, and speed among the three tested models.

- **Model Development**

The validation of the three object detection models by comparing their detection outputs with the evaluation of a human expert. The main goal of the model validation is to determine whether the predictions of YOLOv8, RetinaNet, and Faster R-CNN are statistically different or similar to expert judgment in identifying visible symptoms of Fowlpox in chickens. The comparison was done using a paired samples t-test and supported by descriptive statistics and visual graphs.

The validation datasets included 100 images including both healthy and infected fowlpox chickens, each marked by an expert with 1 if the image shows syntoms of fowlpox and 0 if the image is healthy.

Table 12. Descriptive Statistics Results

	N	Mean	Median	SD	SE
EXPERT	100	0.510	1.00	0.502	0.0502
YOLOv8	100	0.520	1.00	0.502	0.0502
RetinaNet	100	0.770	1.00	0.423	0.0423
Faster R-CNN	100	0.540	1.00	0.501	0.0501

The table above shows the descriptive statistics support the t-test results. The mean score of the expert is 0.51, which is very close to YOLO (0.52) and Faster R-CNN (0.54). These small differences explain why both models did not show a significant difference in the t-test. In contrast, RetinaNet has a mean of 0.77, which is much higher than the expert's score, showing inconsistency. The standard deviation values also show that YOLO and Faster R-CNN are more consistent, while RetinaNet has more variation in its predictions.

Table 13. Results of the Paired Sample T-Test Between Poultry Expert and Model Evaluations.

	Statistic	df	p	Mean Difference	SE Difference	Effective Size (Cohen's d)
YOLO	-0.300	99	0.765	-0.0100	0.0333	-0.0300
RetinaNet	-5.366	99	<.001	-0.2600	0.0485	-0.5366
Faster R-CNN	0.773	99	0.441	-0.0300	0.0388	-0.0773

Table 13 shows the comparison between the

expert's evaluation and the three models (YOLO, RetinaNet, and Faster R-CNN) using a paired sample t-test. The results show that both YOLO ($p = 0.765$) and Faster R-CNN ($p = 0.441$) have p-values higher than 0.05. This means that their outputs are not significantly different from the expert's evaluation. They gave results very close to what the expert observed. However, RetinaNet ($p < 0.001$) has a p-value lower than 0.05, which means there is a significant difference between its output and the expert's evaluation. This indicates that RetinaNet did not match the expert's performance and was less accurate in detecting Fowlpox symptoms.

The results of the paired samples t-test revealed no statistically significant difference between the poultry expert's evaluation and the model's predictions. This indicates that the performance of these models closely aligns with expert judgment, reflecting a strong level of consistency and agreement. The findings suggest that the models are capable of accurately identifying chickens showing visible signs of Fowlpox, and that their predictions are not due to random chance. As a result, the expert validation confirms the reliability and effectiveness of the proposed models, particularly YOLO and Faster R-CNN, in detecting Fowlpox-infected chickens based on visual symptoms.

IV. SUMMARY, CONCLUSIONS AND RECOMMENDATION

Summary of the Study

Fowlpox is one of the most common viral diseases that affect chickens, characterized by wart-like lesions on the comb, wattles, and other unfeathered parts of the body. The disease can lead to reduced productivity, weight loss, and even death if not detected early. Traditionally, farmers rely on manual observation to identify infected chickens, which are often inaccurate and time-consuming. To address this issue, this study was conducted to develop and compare three deep learning models that can automatically detect Fowlpox disease in chickens through images.

The dataset used in this study consisted of 639 raw images of chickens collected from various online sources, including both healthy and Fowlpox-infected samples. Each image underwent manual annotation to mark visible lesions using the Roboflow platform. The images were then preprocessed and augmented to enhance quality and variation. The dataset was divided into 75% for training, 12% for validation, and 13% for testing to ensure a balanced evaluation of model performance and a total of 1275.

The study involved the development and evaluation of three object detection algorithms: YOLOv8, Faster R-CNN, and RetinaNet. Each model was trained using the same dataset and compared based on their precision, recall, f1 score and mean average precision (mAP@50) and (mAP@50-95). The YOLOv8 algorithm, known for its real-time detection capability and high accuracy, achieved the best results with an mAP@50 of 0.98%, precision of 0.95%, recall of 0.97% and f1 score of 0.96% Meanwhile, Faster R-CNN and RetinaNet also demonstrated reliable performance but were slightly less efficient in detecting small or subtle lesions compared to YOLOv8.

The findings indicate that YOLOv8 was the most effective in identifying Fowlpox lesions, showing high detection accuracy and faster processing speed. These results prove that deep learning algorithms, particularly object detection models, can be applied successfully in the early detection of chicken diseases.

Conclusions

This section presents the conclusions derived from the research findings based on the study's objectives. Each objective is addressed to confirm its achievement and relevance to the overall research goal.

1. **Prepare and augment a labeled dataset of fowlpox-infected chickens for model training and testing**.

The study successfully prepared and augmented a labeled dataset of fowlpox-infected and healthy chickens for model training and testing. A total of 639 raw images were collected and carefully labeled with bounding boxes to identify visible fowlpox lesions. The dataset was then divided into training, validation, and testing subsets to ensure proper model evaluation and prevent data bias.

To improve the quality and diversity of the dataset, several augmentation techniques were applied, including horizontal flipping, rotation, brightness and contrast adjustments, mosaic combinations, and conversion to grayscale. The inclusion of grayscale transformation helped

the model focus on texture and shape patterns of lesions rather than color alone, enhancing its ability to detect infections even under poor lighting conditions.

Overall, the preparation and augmentation process successfully produced a balanced, high-quality dataset suitable for robust model training. The resulting dataset provided a strong foundation for object detection algorithms to accurately identify and classify fowlpox-infected chickens, contributing to a more reliable and efficient disease detection system.

2. **Implement and train selected object detection algorithms, namely YOLO, RetinaNet, and Faster R-CNN, for detecting chicken disease.**

The study successfully implemented and trained three state-of-the-art object detection algorithms YOLO, RetinaNet, and Faster R-CNN using the prepared dataset. Each algorithm was carefully configured and optimized for detecting visible symptoms of Fowlpox on chicken skin. YOLO was chosen for its fast and efficient real-time detection capabilities, RetinaNet for its effective handling of class imbalance through the use of focal loss, and Faster R-CNN for its high detection accuracy due to its two-stage detection approach. The training process involved tuning various hyperparameters such as learning rate, batch size, and number of epochs to achieve the best possible performance for each model. After training, all three algorithms were able to successfully detect both healthy and infected chickens, proving the potential of image-based of chicken disease detection in poultry disease monitoring and analysis.

3. **Evaluate and compare the performance of the implemented algorithms using standard metrics such as mAP, precision, recall, and F1-score.**

The evaluation results showed that all three models achieved good performance, but YOLOv8 outperformed RetinaNet and Faster R-CNN in almost all metrics. The YOLOv8 model achieved a Precision of 0.95, Recall of 0.97, F1-Score of 0.96, mAP@50 of 0.98, and mAP@50–95 of 0.82. These values indicate that the model was able to detect and classify Fowlpox-infected and healthy chickens with very high accuracy and consistency. The high precision means that most of YOLOv8's detections were correct, while the high recall shows that it successfully identified nearly all infected chickens. The high mAP values further confirm YOLOv8's ability to precisely localize disease-affected areas in images. Compared to RetinaNet and Faster R-CNN, YOLOv8 achieved superior results in both accuracy and speed, making it the most effective model for this study.

4. **Identify the most effective object detection algorithm for detection of chicken disease based on the comparative analysis.**

Based on the comparative analysis, YOLOv8 was identified as the most effective and reliable object detection algorithm for detecting Fowlpox disease in chickens. With the highest mAP, precision, recall, and F1-score among all tested models, YOLOv8 demonstrated remarkable performance in both identifying and localizing Fowlpox lesions. Its real-time detection capability also makes it practical for on-site poultry monitoring, where quick and accurate results are necessary. While RetinaNet and Faster R-CNN performed well, their detection speed and overall accuracy were slightly lower compared to YOLOv8. Therefore, YOLOv8 proved to be the most efficient and accurate model for the image-based detection of Fowlpox disease.

In conclusion, this study successfully achieved all its objectives and demonstrated that YOLOv8 is the most effective deep learning-based object detection algorithm for identifying Fowlpox disease in chickens. The preparation and augmentation of the dataset significantly improved model training, while the comparison among YOLOv8, RetinaNet, and Faster R-CNN provided valuable insights into the performance of different object detection frameworks. With its outstanding results Precision (0.95), Recall (0.97), F1-Score (0.96), mAP@50 (0.98), and mAP@50–95 (0.82). YOLOv8 demonstrated superior detection accuracy and efficiency. The findings of this study highlight the potential of integrating YOLOv8 into automated poultry monitoring systems for early detection of Fowlpox disease, helping farmers reduce losses, improve poultry health management, and promote more sustainable poultry farming practices.

Recommendations

Based on the results and findings of the study titled "Comparative Analysis of Object Detection Algorithms for Image-Based

Detection of Chicken Fowlpox Disease," the following recommendations are proposed:

1.Implement the Model in Poultry Farms

It is recommended to apply the developed detection system in actual poultry farms to evaluate its effectiveness in real-world conditions. This will help determine the model's practicality and accuracy in identifying Fowlpox symptoms among live chickens.

2.Expand the Dataset

Future researchers should collect a larger and more diverse dataset consisting of chicken images captured under various lighting conditions, environments, and camera angles. A broader dataset will enhance the model's learning capability and overall detection performance.

3. Include Additional Chicken Diseases

The detection system can be improved by training it to identify other common chicken diseases in addition to Fowlpox. This enhancement will make the system more useful for comprehensive poultry health monitoring.

4.Develop Real-Time Monitoring Capability

It is suggested to integrate real-time detection features into the system so that disease identification can occur instantly through live video feeds or camera surveillance. This would allow faster response and early disease prevention.

5. Utilize More Advanced Deep Learning Algorithms

Future studies should explore the use of newer and more advanced deep learning algorithms, such as YOLOv11 or transformer-based object detection models, to further improve detection speed, accuracy, and efficiency.

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